

WHAT MATTERS FOR REPRESENTATION ALIGNMENT: GLOBAL INFORMATION OR SPATIAL STRUCTURE?

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ABSTRACT

Representation alignment (REPA) guides generative training by distilling representations from a strong, pretrained vision encoder into intermediate diffusion features. This paper asks a single, fundamental question and answers it in three steps: what aspect of the target representation actually matters for generation, its *global* semantic information (e.g., measured by ImageNet-1K accuracy) or its spatial structure (i.e., pairwise cosine similarity between patch tokens)? The prevailing wisdom holds that stronger global semantic performance yields a better target representation for generation. **Our first contribution** overturns this wisdom: through a large-scale empirical analysis spanning 27 vision encoders and three model scales, we show that spatial structure, and not global performance, is what drives the generation performance of a target representation. **Our second contribution** makes this insight actionable: we introduce a simple, fast-to-compute Spatial Structure Metric that correlates far more strongly with generation quality than ImageNet-1K accuracy. **Our third contribution** turns the insight into a method: two straightforward modifications, in under 4 lines of code, that specifically accentuate the transfer of *spatial* information, replacing the standard MLP projection layer in REPA with a simple convolution layer and adding a spatial normalization layer for the external representation. The result, termed iREPA, consistently improves the convergence speed of REPA across a diverse set of vision encoders, model sizes, and training variants (REPA, REPA-E, MeanFlow, JiT, etc.). Together, these contributions motivate revisiting the fundamental working mechanism of representation alignment and how it can be leveraged for improved training of generative models.

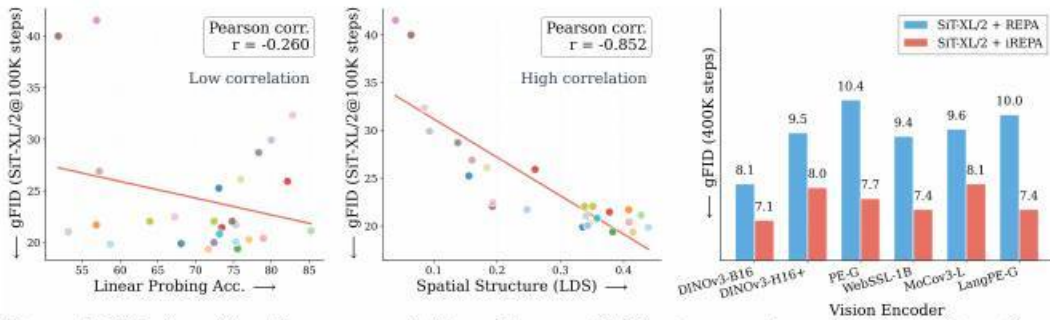


Figure 1: **What matters for representation alignment?** The two panels preview two of our three contributions. *Left (Contribution 1–2)*: a correlation analysis across 27 diverse vision encoders shows that, contrary to the prevailing wisdom, spatial structure — not global performance (measured by linear probing accuracy) — drives the generation performance of a target representation. *Right (Contribution 3)*: acting on this insight, two simple modifications that accentuate the transfer of spatial features from the target representation to the diffusion model consistently improve the convergence speed of REPA across diverse settings.



Figure 2: **Motivating examples for Contribution 1 — spatial structure matters.** These comparisons expose the inverse relationship between ImageNet accuracy and generation quality that motivates our study. *Left:* PE-G, despite much higher validation accuracy (82.8% vs. 53.1%), generates worse than SpatialPE-B (Bolya et al., 2025). *Right:* likewise, WebSSL-1B (Fan et al., 2025), with much higher global performance (76.0% vs. 53.1%), generates worse. **Spatial self-similarity:** spatial structure predicts generation quality where global performance does not, previewing Contribution 2. See §3 for the spatial structure metric. All results reported at 100K using SiT-XL/2 and REPA.

1 INTRODUCTION

Representation alignment has emerged as a powerful technique for accelerating the training of diffusion transformers (Ma et al., 2024; Peebles & Xie, 2023). By aligning internal diffusion representations with pretrained self-supervised visual encoders, recent methods have demonstrated significant improvements in the convergence speed and final performance (Yu et al., 2024; Leng et al., 2025a). Yet despite these empirical successes, we still lack a clear understanding of the precise mechanism through which self-supervised features enhance diffusion model training. This paper is organized around one fundamental question: is the improvement primarily driven by incorporating better global semantic information, as commonly measured through linear probing performance, or does it instead stem from better capturing spatial structure, characterized by the relationships between patch token representations? Answering this question is the through-line of our work, and each of our contributions builds directly on the answer.

Getting this answer right matters for practice, not just for understanding: it directly determines how one should select the target representation and maximize its benefits. Today, the prevailing assumption is that encoder quality for representation alignment tracks ImageNet-1K validation accuracy, a proxy for global semantic understanding (Oquab et al., 2024; Chen et al., 2021). In other words, target representations with better ImageNet performance are hypothesized to lead to better generation (Yu et al., 2024). Consistent with this view, increases in linear probing accuracy of diffusion features are routinely cited as evidence for the effectiveness of representation alignment, reinforcing global semantic information as the presumed primary driver of improvement. The following quote from REPA (Yu et al., 2024) crisply captures this current understanding, and it is exactly the belief our first contribution puts to the test:

“When a diffusion transformer is aligned with a pretrained encoder that offers more semantically meaningful representations (i.e., better linear probing results), the model not only captures better semantics but also exhibits enhanced generation performance, as reflected by improved validation accuracy with linear probing and lower FID scores.”

In this paper, we challenge this conventional wisdom by systematically investigating what truly drives representation alignment: global semantic information or spatial structure? Through large-scale empirical analysis across diverse vision encoders, including recent large vision foundation models such as WebSSL (Fan et al., 2025), DINOv3 (Siméoni et al., 2025), perceptual encoders (Bolya et al., 2025), and C-RADIO (Heinrich et al., 2024), we arrive at three surprising findings. We state each finding up front below, and the three together map onto the three contributions that structure the rest of the paper: an empirical study that reframes the question, a metric that answers it quantitatively, and a method that exploits the answer.

Finding 1 (Contribution 1): Higher validation accuracy does not imply a better representation for generation. Contrary to the prevailing assumption, vision encoders with higher global semantic performance, measured by ImageNet-1K linear probing accuracy, often underperform for generation. For instance, consider PE-Spatial-B, a spatially-tuned model derived from PE-Core-G (Bolya et al.,

2025). Although PE-Spatial-B has far worse patch-token validation accuracy¹ than PE-Core-G (53.1% vs. 82.8%), it yields *better* generation with REPA (Figure 2). Likewise, WebSSL-1B (Fan et al., 2025) enjoys much stronger global performance (76.0% vs. 53.1%) yet produces worse generation. As our first contribution (§2), we show through a large-scale study that this inversion is not anecdotal but holds across a wide range of target representations, pointing to a fundamental principle in how representation alignment benefits diffusion training.

Finding 2 (Contribution 2): Spatial structure, not global information, determines generation performance. To turn this insight into a measurable quantity, our second contribution introduces several straightforward metrics for the spatial self-similarity structure (Shechtman & Irani, 2007) between patch tokens (§3). Using these metrics, we perform a large-scale analysis correlating generation FID under REPA against linear probing accuracy and against spatial structure, spanning 27 vision encoders and 3 model sizes (SiT-B, SiT-L, SiT-XL). The verdict is decisive: every spatial structure metric correlates strongly with generation FID (Pearson $|r| > 0.852$), far surpassing the predictive power of ImageNet-1K validation accuracy ($|r| = 0.26$). Two further experiments sharpen the point that spatial, rather than global, information is what matters: external representations carrying very little global information still deliver significant REPA gains. SAM2, for example, outperforms encoders with $\sim 60\%$ higher ImageNet accuracy (Fig. 3), and even classical spatial features such as SIFT (Lowe, 1999) and HOG (Dalal & Triggs, 2005), while not the best, achieve solid gains with representation alignment (§3).

Finding 3 (Contribution 3): Accentuating spatial features improves convergence. Our third contribution puts the analysis to work: two straightforward modifications (<4 lines of code) that specifically accentuate the transfer of “spatial” information from the target representation to the diffusion model. (1) We add a spatial regularization layer that boosts the spatial contrast of the target representations. (2) We show that the standard MLP-based projection layer (used to map diffusion features to target-representation dimensions) causes a loss of spatial information (Figure 6a), and replace it with a simple convolution layer. The payoff is broad and consistent: our method, termed iREPA, reliably speeds up REPA convergence across diverse encoders, model sizes, and training recipes, *e.g.*, REPA-E (Leng et al., 2025a) and MeanFlow (Geng et al., 2025) w/ REPA.

In summary, this paper makes three contributions, one per finding above:

- **Contribution 1 (Empirical study).** We perform a large-scale empirical analysis across 27 vision encoders and 3 model scales, establishing that spatial structure, and not global semantic information, drives the effectiveness of representation alignment.
- **Contribution 2 (Metric).** We introduce a straightforward, fast-to-compute Spatial Structure Metric (SSM) that correlates substantially more strongly with downstream FID than linear probing accuracy, and use it to explain previously unexplained REPA trends.
- **Contribution 3 (Method).** We propose simple modifications (iREPA, <4 lines of code) that accentuate the transfer of spatial information from the target representation to diffusion features, yielding consistent convergence-speed improvements over REPA across target representations, model architectures, and training-recipe variants (REPA, REPA-E, MeanFlow w/ REPA, JiT w/ REPA, etc.).

2 MOTIVATION: GLOBAL INFORMATION MATTERS LESS

This section presents our first contribution: an empirical study establishing that a target representation with higher global performance (ImageNet-1K accuracy) does not imply better generation performance with REPA. We build the case through four complementary observations, each isolating a different way in which global performance fails as a predictor. Each observation is left partly unexplained here on purpose; §3 shows that all four are cleanly accounted for once we measure the spatial structure of the target representation (our second contribution).

Observation 1: Recent vision encoders. We first examine several recent vision encoders, including Perceptual Encoders (Bolya et al., 2025), WebSSL (Fan et al., 2025), and DINOv3 (Siméoni et al., 2025). Consider PE-Spatial-B (80M), a small spatially tuned model derived from PE-Core-G (1.88B)

¹Similar to REPA (Yu et al., 2024), we use linear probing accuracy on patch tokens to measure global semantic performance of external representation as only patch tokens are used for representation alignment.

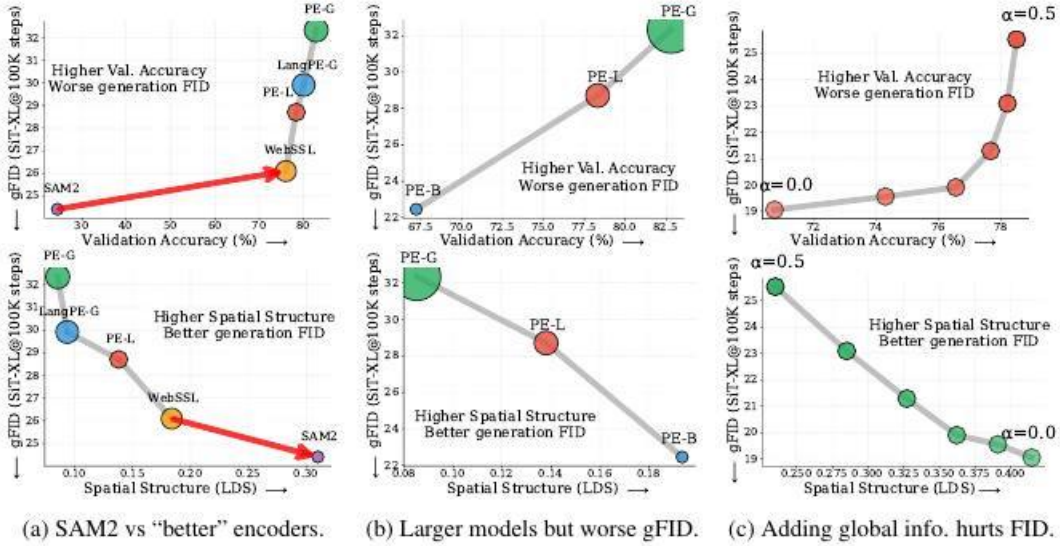


Figure 3: **Contribution 1 in one figure: higher global information does not imply better REPA performance.** **Top row (the puzzle):** three trends in which global performance fails to track generation FID under REPA. (a) SAM2-S, with validation accuracy of only 24.7%, generates better with REPA than models with $\sim 60\%$ higher validation accuracy. (b) Larger encoders within the same family can have higher validation accuracy yet worse generation. (c) Adding global information to patch tokens via the CLS token raises global performance but hurts generation. **Bottom row (the resolution, Contribution 2):** the same trends line up cleanly with spatial structure, which is the better indicator for generation. See §3 for the large-scale analysis across spatial structure metrics. All results are reported w/o classifier free guidance, SiT-XL/2 w/ REPA and 250 NFE (Yu et al., 2024) for inference.

(Bolya et al., 2025). As Figure 2 shows, PE-Core-G achieves much higher ImageNet-1K accuracy (82.8% vs. 53.1%) yet performs *worse* as the target representation for REPA (FID 32.3 vs. 21.0). The same inversion recurs with WebSSL-1B (1.2B): despite far higher ImageNet-1K accuracy (76.0% vs. 53.1%), it too yields worse generation under REPA (FID 26.1 vs. 21.0).

Observation 2: SAM2 outperforms vision encoders with much higher ImageNet-1K accuracy. To test how far global information can be stripped away, we analyze the SAM2-S vision encoder (46M) (Ravi et al., 2024) — a small model carrying very little global information, with validation accuracy of only 24.1% (Fig. 3a). The result runs directly against the prevailing wisdom: used for REPA, SAM2-S attains better FID than encoders with dramatically higher ImageNet-1K accuracy, e.g., PE-Core-G (82.8%).

Observation 3: Larger models within the same encoder family can have similar or worse generation performance. A common expectation is that larger models within an encoder family carry better representations (as measured by ImageNet-1K accuracy). For representation alignment, however, larger variants often deliver similar (DINOv2) or even *worse* (PE, Cradio) generation performance (Fig. 3b). Notably, (Yu et al., 2024) report a similar effect for DINOv2 and attribute it to “we hypothesize is due to all DINOv2 models being distilled from the DINOv2-g model and thus sharing similar representations.” We show in §3 that spatial structure offers a cleaner and more general explanation.

Observation 4: Adding more global information can hurt generation. Finally, we test the converse directly: does *injecting* global semantics into a fixed representation help? We run controlled experiments that mix the CLS token into local patch tokens (with DINOv2 as encoder). The CLS token mixing operation is $\mathbf{p}_i^{\text{new}} = \mathbf{p}_i + \alpha \cdot \mathbf{c}$, where $\mathbf{p}_i$ denotes the i^{th} patch token, $\mathbf{c}$ the CLS token, and $\alpha \in [0, 0.5]$ controls the mixing strength. As Figure 3c shows, increasing α from 0 to 0.5 raises linear probing accuracy monotonically from 70.7% to 78.5% — yet generation quality deteriorates sharply, with FID worsening from 19.2 at $\alpha = 0$ to 25.4 at $\alpha = 0.5$. More global information, worse generation.

Taken together, these four observations establish our first contribution: the global performance of a representation is not a reliable indicator of its REPA performance. The natural next question is what *is* the right indicator — which we answer in §3, where spatial structure emerges as the missing signal.

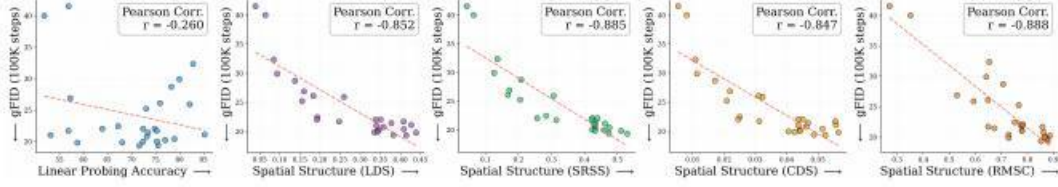


Figure 4: Core evidence for Contribution 2: spatial structure correlates far more strongly with generation quality than linear probing. Correlation analysis across 27 diverse vision encoders (SiT-XL/2, REPA). Linear probing correlates only weakly with FID (Pearson $|r| = 0.260$), whereas every spatial structure metric — LDS ($|r| = 0.852$), SRSS ($|r| = 0.885$), CDS ($|r| = 0.847$), and RMSC ($|r| = 0.888$) — is far more predictive of generation performance. See Fig. 10 for detailed plots with encoder labels.

- **Finding 1.** Better global semantic information (e.g., ImageNet-1K accuracy) does not imply a better representation for generation.

3 SPATIAL STRUCTURE MATTERS MORE

Having shown that global performance fails to predict REPA quality, we now deliver our second contribution: a spatial signal that succeeds. We hypothesize that spatial structure, rather than global information, drives the generation performance of a target representation, and we make this hypothesis testable by introducing several straightforward metrics of spatial self-similarity structure (Shechtman & Irani, 2007). The section then supplies two pieces of evidence for the contribution: (i) every spatial structure metric correlates far more strongly with gFID than ImageNet-1K accuracy, and (ii) these same metrics cleanly explain each of the previously unexplained observations from §2.

Measuring spatial self-similarity structure. Given an image $\mathcal{I}$ and vision encoder $\mathcal{E}$, we define:

- *Self-similarity.* Let $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ be the extract patch representations, with $T = H \times W$ patches. Let kernel $K_{\mathcal{X}}(\cdot, \cdot) \in \mathbb{R}$ measure appearance self-similarity (Shechtman & Irani, 2007) between patch tokens. Here, we use the cosine kernel $K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2}$.
- *Spatial distance.* Let $\mathcal{P} \in \mathbb{N}^{T \times 2}$ be the spatial location of each of the T tokens in coordinate space and $d(\cdot, \cdot) \in \mathbb{N}$ be the Manhattan distance between pairs of tokens.

We then measure *spatial* self-similarity metric as how self-similarity $K_{\mathcal{X}}$ varies with lattice distance d between patch tokens. Intuitively, larger values indicate stronger spatial organization (closer patches more similar to each other than patches further away). By default, we use a simple correlogram contrast (local vs. distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}, \mathcal{P}) = \mathbb{E}_{d(t, t') \in (0, r_{\text{near}})} K_{\mathcal{X}}(t, t') - \mathbb{E}_{d(t, t') \geq r_{\text{far}}} K_{\mathcal{X}}(t, t').$$

The final spatial self-similarity score (LDS) is the expectation over patch representations $\mathcal{X}$. We set $r_{\text{near}} = r_{\text{far}} = H/2$ here, and found the correlation to be robust to their exact choices. Importantly, our contribution is not tied to any single metric: LDS is only the default. Appendix B gives full details and three alternative spatial metrics (CDS, SRSS, RMSC), all of which are similarly predictive (Fig. 4).

Main result: spatial structure correlates far more strongly with generation performance than linear probing. This is the core quantitative evidence for our second contribution. Across a large-scale correlation analysis over 27 diverse vision encoders (Figure 4), the commonly used linear probing accuracy correlates only very weakly with generation FID (Pearson $|r| = 0.26$), whereas every SSM metric correlates strongly (Pearson $|r| > 0.85$) — a more than threefold gap in predictive power.

The result generalizes across model scales. To rule out a scale-specific artifact, we repeat the analysis at three model scales (SiT-B, SiT-L, SiT-XL) in Figure 5. The gap persists at every scale: linear probing remains weak ($|r| < 0.306$) while spatial structure stays strongly predictive ($|r| > 0.826$).

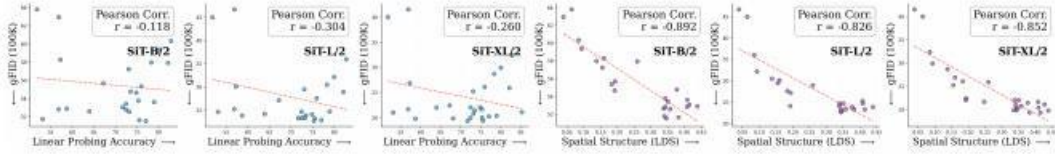


Figure 5: **Contribution 2 holds across model scales.** At every model scale, spatial structure (right) is consistently more correlated with gFID than linear probing (left), showing the metric’s advantage is not scale-specific.

- **Finding 2.** *Spatial structure correlates much higher with generation performance than linear probing. We next use spatial metrics to explain previously unexplained trends.*

The metric explains every previously unexplained trend from §2. Beyond raw correlation, a good indicator should account for the specific inversions that defeated global performance. Here we revisit all four observations from our empirical study and show that spatial structure explains each one, closing the loop opened by our first contribution.

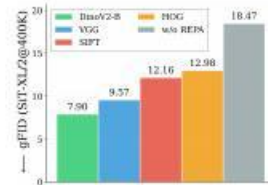
(1) *PE-Spatial-B vs. PE-Core-G:* Figure 2 shows PE-Core-G with much higher ImageNet-1K accuracy (82.8% vs. 53.1%) but worse REPA generation (FID 32.3 vs. 22.0). The spatial view resolves the paradox: as Figure 2 makes clear, despite its lower global performance, PE-Spatial-B has much stronger spatial structure than PE-Core-G, which is precisely what drives its better generation.

(2) *SAM2 outperforms “better” vision encoders:* recall from §2 that SAM2, despite very low ImageNet-1K accuracy (24.1%), beats encoders with $\sim 60\%$ higher accuracy. Fig. 3a shows these gains follow directly from SAM2’s superior spatial structure.

(3) *Larger models in the same encoder family underperform:* Fig. 3b shows that larger family members, though higher in ImageNet-1K accuracy, can have weaker spatial structure and hence worse REPA generation. This also aligns with Yu et al. (2024), who report that larger models can yield similar or worse generation.

(4) *Adding global information to patch tokens via the CLS token hurts generation:* Figure 3c shows that mixing in the CLS token raises global performance but suppresses spatial contrast among patch tokens, so that patch tokens become highly similar to otherwise unrelated tokens (e.g., foreground object to background). The resulting loss of spatial structure is what degrades generation. In every case, the direction of the spatial-structure metric matches the direction of generation quality, exactly where global performance did not.

A stress test: if spatial structure matters more, can even SIFT or HOG features work for REPA? Surprisingly, yes. While certainly not the best, classical spatial features — SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), and intermediate VGG features (Simonyan & Zisserman, 2014) — all yield gains with REPA. This is arguably the strongest form of our second contribution’s claim: representation alignment can benefit from spatial features alone, with essentially no additional global information.



Do the spatial metrics also explain the gains from REPA itself? Yes. As shown, the introduced spatial metrics account both for the gains from REPA and for those of our improved training recipe (iREPA), which we introduce next in §4 as our third contribution.



4 iREPA: IMPROVING REPRESENTATION ALIGNMENT BY ACCENTUATING WHAT MATTERS

Our third contribution turns the preceding analysis into a concrete, deployable recipe. If spatial structure is what matters, then explicitly accentuating its transfer should improve REPA — and it does. We introduce two straightforward modifications (<4 lines of code in total) to the original REPA recipe, each targeting a distinct way spatial information is otherwise lost in transfer from the teacher (vision encoder) to the student (diffusion transformer). We describe the two modifications below, then validate the combined recipe, iREPA, through an extensive set of experiments.

Modification 1: convolutional projection layer instead of MLP. The standard REPA recipe uses a 3-layer MLP projection to map diffusion feature dimensions to those of the external representation. As Figure 6a shows, this projection is lossy and diminishes the spatial contrast between patch tokens — directly working against the very signal our metric identifies as useful. We therefore replace the MLP with a lightweight convolutional layer (kernel size 3, padding 1) that operates directly on the spatial grid, whose inductive bias naturally preserves local spatial relationships.

Modification 2: spatial normalization layer. Similar to Siméoni et al. (2025), we find that the patch tokens of pretrained vision encoders carry a significant global component, evidenced by the high linear probing scores for the mean of patch tokens (Figure 14). As Figure 6b shows, mixing this global information (the mean of patch tokens) into the patch tokens improves global performance but reduces the spatial contrast between individual tokens, so that tokens (e.g., a foreground object) become highly similar to otherwise unrelated tokens (e.g., background).

The analysis in §3 makes the design choice obvious: we can afford to sacrifice this global component precisely because it is not what helps generation. Doing so sharpens the spatial contrast between patch tokens, strengthening the pairwise-similarity signal that drives REPA. Concretely, we add a simple spatial normalization layer (Ulyanov et al., 2016) to the patch tokens of the target representation:

$$\mathbf{y} = \frac{\mathbf{x} - \gamma \mathbb{E}[\mathbf{x}]}{\sqrt{\text{Var}[\mathbf{x}] + \epsilon}},$$

where $\mathbf{x} \in \mathbb{R}^{B \times T \times D}$ represents the patch tokens, the expectation and variance are computed across the spatial dimension, and $\epsilon = 10^{-6}$ for numerical stability.

Algorithm 1 Summary of key iREPA modifications to standard REPA training

```
# 1. Conv projection instead of MLP
proj_layer = nn.Conv2d(D_in, D_out, kernel_size=3, padding=1)

# 2. Spatial normalization on encoder features [B, T, D]
x = x - gamma * x.mean(dim=1, keepdim=True)
x = x / (x.std(dim=1, keepdim=True) + 1e-6)
```

4.1 EXPERIMENTS

We now validate our third contribution through extensive experiments on ImageNet 256×256 . To make the scope of the claim explicit, we frame the evaluation around three research questions, spanning encoders, scale, and training recipes:

- Can iREPA consistently improve the convergence speed of diffusion transformers over REPA across diverse external representations? (Figure 7, 8, 12, 13, 15)
- Is iREPA scalable in terms of model size and generalize across variations in training settings? (Table 1 [a,b,c], 2, 4, 5, 6, and, Figure 12, 16)
- Does iREPA generalize across more recent representation alignment methods such as REPA-E (Leng et al., 2025a), MeanFlow w/ REPA (Geng et al., 2025)? and pixel-space diffusion models such as JiT w/ REPA (Li & He, 2025b)? (Table 3 [a,b])

Convergence speed (RQ1). We first evaluate convergence across diverse vision encoders (DINOv3-B, WebSSL-1B, PE-Core-G, CLIP-L, MoCov3, PE-Lang-G) and model sizes (SiT-XL/2 and SiT-B/2). As Fig. 7 shows, iREPA consistently reaches a given FID faster than baseline REPA, across every combination of target representation and model size we tried — the central claim of our third contribution.

Generality across target representations (RQ1). We next probe how broadly the gains hold across vision encoders (Fig. 8 and Table 4). The improvement is universal: iREPA improves generation quality for every vision encoder tested, and this holds across all 27 encoders reported in Appendix C, E.

Scaling with encoder size (RQ2). Turning to scalability, Table 1a reports iREPA across encoder sizes PE-B (90M), PE-L (320M), and PE-G (1.88B). iREPA improves performance at every encoder size, and — notably — the relative gain *grows* with encoder size (22.2% for PE-B, 38.8% for PE-L,

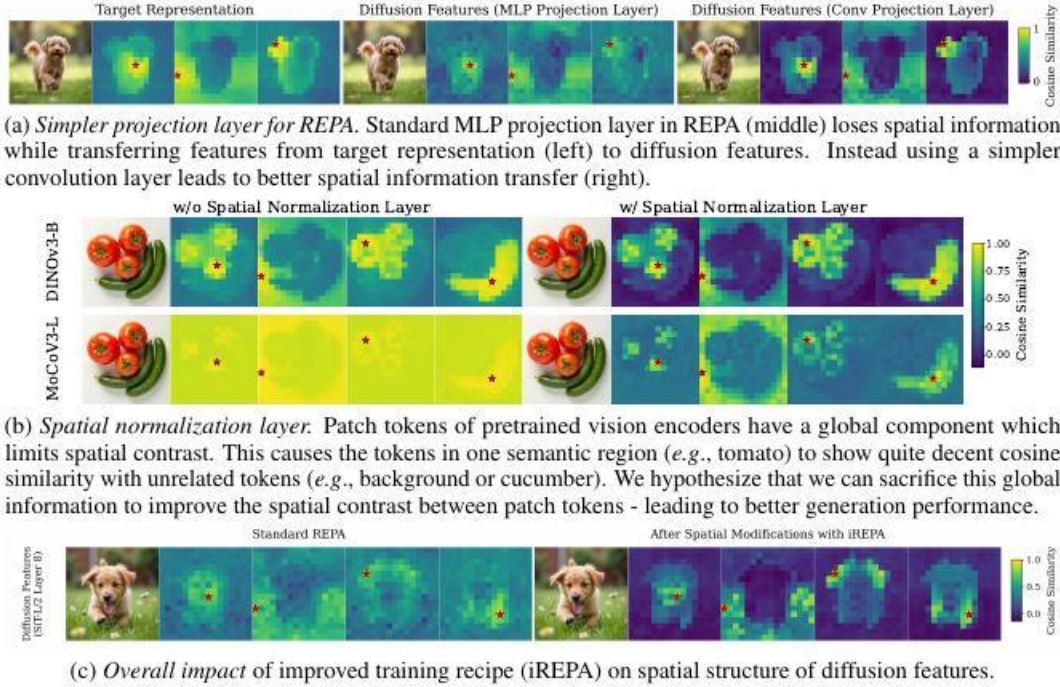


Figure 6: **The two modifications behind Contribution 3 (iREPA), and why each is needed.** Both modifications enhance spatial feature transfer from the target representation to diffusion features: (a) a convolutional projection layer avoids the spatial information loss of the standard MLP, and (b) a spatial normalization layer removes the global component that suppresses spatial contrast. All results reported with SiTL/2 with REPA at 100K.

Enc. Size	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$	Model Size	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$	Aln. Depth	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$
PE-B (90M)	59.6	22.5	6.23	0.63	0.60	SiT-B	27.5	49.50	7.00	0.46	0.59	Layer 4	28.1	49.0	6.83	0.46	0.60
+iREPA	73.3	17.5	6.04	0.66	0.62	+iREPA	34.1	43.37	7.87	0.50	0.60	+iREPA	41.7	36.0	6.61	0.52	0.62
PE-L (320M)	48.6	28.7	6.53	0.59	0.60	SiT-L	55.7	24.10	6.25	0.62	0.60	Layer 6	26.9	50.6	7.00	0.46	0.59
+iREPA	76.3	17.6	6.36	0.65	0.61	+iREPA	66.9	20.28	6.14	0.63	0.62	+iREPA	42.4	35.3	6.90	0.53	0.62
PE-G (1.88B)	42.7	32.3	6.70	0.57	0.59	SiT-XL	70.3	19.06	5.83	0.65	0.61	Layer 8	24.7	54.8	7.35	0.44	0.58
+iREPA	70.8	19.5	6.35	0.64	0.61	+iREPA	77.9	16.96	6.26	0.66	0.61	+iREPA	38.3	38.9	7.19	0.51	0.62

(a) Vision Encoder size

(b) Model size

(c) Encoder depth (SiT-B/2)

Table 1: **Contribution 3 scales (RQ2): consistent gains across training settings.** iREPA improves over baseline REPA across encoder size (a), model size (b), and alignment depth (c), with relative gains that tend to grow with scale. Unless otherwise specified, all results use DINOv2 as encoder, SiTXL/2, 100K steps, and vanilla-REPA as baseline.

39.6% for PE-G), so the contribution becomes more valuable exactly where baseline REPA struggles most.

Scaling with model size (RQ2). Table 1b examines the complementary axis, model scale. Here too iREPA consistently improves performance, and the relative gain again increases with model size — evidence that our third contribution scales up rather than washing out on larger diffusion transformers.

Robustness to alignment depth (RQ2). Table 1c varies the alignment depth (SiT-B/2, 100K iterations, DINOv3-B as target). The gains over baseline REPA hold at every depth, showing the contribution is not sensitive to this design choice.

Ablation: both modifications contribute (RQ2). To confirm that each of the two modifications earns its place, Table 2 ablates them. Both the spatial normalization layer and the convolutional projection layer individually improve generation quality over baseline REPA, and the two are complementary: the best results come from using them together.

Generality across training recipes (RQ3). Finally, we test whether the contribution transfers beyond vanilla REPA to more recent recipes such as REPA-E (Leng et al., 2025a) and MeanFlow w/ REPA

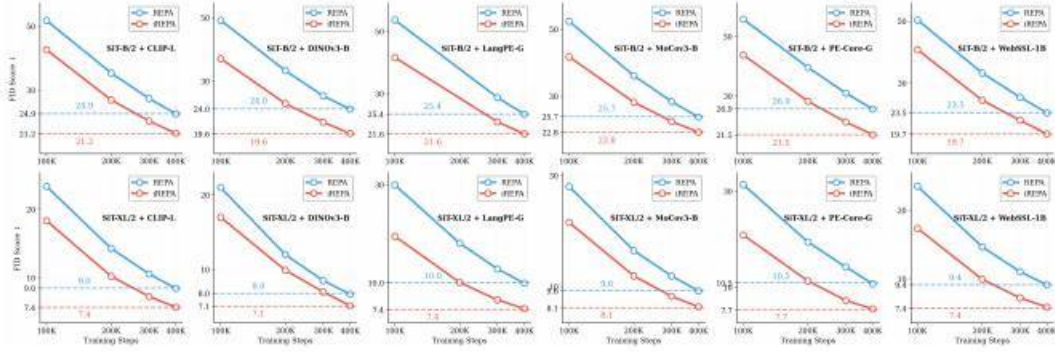


Figure 7: **Contribution 3, at a glance: accentuating spatial features consistently improves convergence speed.** Across encoders and model sizes, iREPA reaches a given FID faster than baseline REPA.

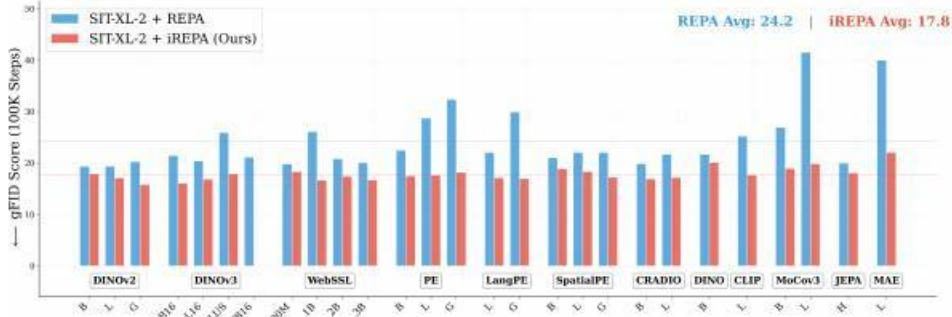


Figure 8: **Contribution 3 generalizes across all 27 target representations.** Accentuating the transfer of spatial features from the target representation to diffusion features (iREPA) consistently improves convergence speed for every encoder. See Appendix C for more results across diverse settings.

(Geng et al., 2025). Table 3 shows that the spatial improvements carry over intact, delivering convergence gains under each recipe.

Robustness to classifier-free guidance (RQ1). Table 4 reports iREPA with CFG across different vision encoders at 400K training iterations. The convergence advantage holds regardless of guidance: across encoders, iREPA converges faster both with and without classifier-free guidance.

Extension to pixel-space diffusion, JiT (RQ3). As a final generalization test, we apply iREPA to pixel-space diffusion models such as JiT-B (Li & He, 2025b). Figure 9 compares REPA and iREPA under JiT (Li & He, 2025b): accentuating the transfer of spatial information again yields consistently faster convergence across vision encoders (e.g., DINOv2, DINOv3, PE, etc.), extending our third contribution beyond latent-space diffusion.

- **Finding 3.** Accentuating transfer of spatial structure from target representation to diffusion features helps boost convergence speed with representation alignment (REPA).

5 RELATED WORK

We discuss the most relevant related work here, positioning each of our three contributions against it, and provide a detailed discussion in Appendix H.

Representation alignment for generation. Many recent works use external representations to improve diffusion model training (Pernias et al., 2023; Fuest et al., 2024). Notably, (Yu et al., 2024;

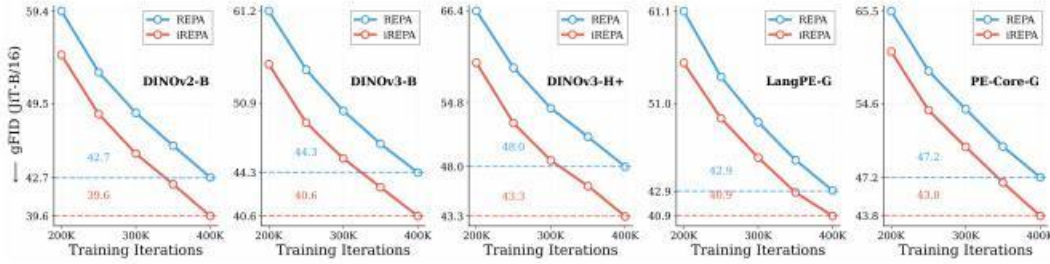


Figure 9: **Contribution 3 extends to pixel-space diffusion (JiT).** FID convergence curves comparing REPA vs. iREPA under pixel-space JiT (Li & He, 2025b) across vision encoders. Accentuating the transfer of spatial structure again consistently speeds up convergence, showing the method is not confined to latent-space diffusion. All results are reported with JiTB/16 (Li & He, 2025b), 256 batch size and without classifier-free guidance. Refer to Table 10 for further results.

Method	DINOv2-B			DINOv3-B			WebSSL-1B			PE-Core-G		
	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓
Baseline REPA	19.06	70.3	5.83	21.47	63.4	6.19	26.10	53.0	6.90	32.35	42.7	6.70
iREPA (w/o spatial norm)	18.52	73.3	6.11	17.76	74.7	5.81	21.17	64.6	6.27	24.97	57.4	6.21
iREPA (w/o conv proj)	17.66	72.8	6.03	18.28	70.8	6.18	18.44	71.0	6.22	21.72	61.5	6.26
iREPA (full)	16.96	77.9	6.26	16.26	78.8	6.14	16.66	77.5	6.18	18.19	75.0	6.03

Table 2: **Ablation: both modifications behind Contribution 3 pull their weight.** Across encoders, the spatial normalization layer and the convolutional projection layer (§4) each give significant convergence-speed gains, and the two are complementary — the best results come from using both. All results reported at 100K steps and SiT-XL/2.

Yao & Wang, 2025; Leng et al., 2025a;b; Kouzelis et al., 2025) show that significant gains follow from aligning internal diffusion features with clean image features from a pretrained vision encoder, and (Zhang et al., 2025; Wu et al., 2025) extend the idea to video and 3D generation, while (Ma et al., 2025a) applies it to unified models. What all of this work leaves open — and what our first two contributions address — is *why* it works: the precise mechanism through which self-supervised features help remains unclear. We identify that mechanism (spatial structure) and, as our third contribution, turn it into an improved training recipe.

Spatial vs. global information tradeoff in pretrained vision encoders. A parallel line of work studies the tradeoff between global and spatial information in pretrained encoders (Bolya et al., 2025; Siméoni et al., 2025). (Siméoni et al., 2025) shows that continued self-supervised training can increase the similarity between the global CLS token and patch tokens, hurting dense spatial tasks, and (Bolya et al., 2025; Heinrich et al., 2024) train spatial-tuned models specifically for such tasks. Our contribution to this discussion is to show that the same spatial-over-global preference holds for *generation*: for REPA, the spatial structure of an encoder matters more than its global information. We hope this reframing motivates future work on selecting and training external representations for generation.

6 CONCLUSION

This paper asked what truly drives the effectiveness of representation alignment — global information or the spatial structure of the target representation — and answered it through three contributions. *First*, a large-scale empirical study across 27 encoders and 3 scales overturns the prevailing wisdom: spatial structure, not global information, drives REPA’s effectiveness. *Second*, we introduce a simple, fast-to-compute Spatial Structure Metric that correlates far more strongly with generation quality than linear probing accuracy and explains previously unexplained REPA trends. *Third*, we turn this understanding into iREPA, two modifications (in under 4 lines of code) that accentuate spatial transfer and consistently improve REPA’s convergence speed across diverse vision encoders, model sizes, and training recipes. Together, these results argue for revisiting the fundamental working mechanism of representation alignment, and we hope they motivate future work on how to better leverage it for training generative models.

Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
WebSSL-1B	52.8	26.5	5.20	0.620	0.585
+iREPA-E	87.0	13.2	5.28	0.699	0.598
PE-G	50.9	25.9	5.68	0.612	0.576
+iREPA-E	80.0	16.4	5.40	0.667	0.616
DINOv3-B	82.2	14.4	4.68	0.694	0.596
+iREPA-E	93.6	11.7	4.57	0.703	0.613
DINOv2-B	87.5	12.9	5.40	0.708	0.586
+iREPA-E	91.3	12.1	4.86	0.712	0.602

(a) REPA-E (SiT-XL/2)

Encoder	w/o CFG				w/ CFG			
	4 NFE		1 NFE		4 NFE		1 NFE	
	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$
WebSSL-1B	27.22	51.40	24.14	58.67	87.85	16.59	69.06	23.74
+iREPA	31.48	45.67	27.33	55.70	100.74	13.89	78.67	20.69
DINOv3-B	28.36	49.58	25.47	57.01	93.30	15.56	72.41	22.55
+iREPA	33.63	44.52	29.67	53.75	124.54	11.05	98.93	17.32
DINOv2-B	28.77	48.87	25.43	56.63	94.40	15.28	71.95	22.17
+iREPA	33.63	44.52	29.67	53.75	111.44	12.59	90.25	18.62

(b) MeanFlow w/ REPA (SiT-B/2)

Table 3: **Contribution 3 transfers across training recipes (RQ3).** Adding our spatial improvements on top of REPA-E (a) and MeanFlow with REPA (b) yields consistent gains, showing the method is recipe-agnostic. For MeanFlow, we report results at 1 and 4 NFEs. CFG value of 2.0 is used for MeanFlow. All results are reported at 100K training iterations.

Vision Encoder	Steps	w/o CFG					w/ CFG				
		IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	100K	69.20	19.3	5.89	0.64	0.61	157.2	6.35	5.91	0.769	0.536
+iREPA	100K	77.92	16.9	6.26	0.66	0.61	179.3	5.15	6.23	0.783	0.544
DINOv2-B	400K	127.4	7.76	5.06	0.70	0.66	263.0	1.98	4.60	0.799	0.610
+iREPA	400K	128.6	7.52	4.89	0.71	0.65	268.8	1.93	4.59	0.799	0.600
DINOv3-B	100K	63.64	21.4	6.14	0.63	0.60	144.0	7.57	6.09	0.762	0.526
+iREPA	100K	78.79	16.2	6.14	0.66	0.61	181.9	4.87	6.10	0.780	0.547
DINOv3-B	400K	126.7	8.10	5.06	0.70	0.66	261.2	1.99	4.58	0.799	0.609
+iREPA	400K	132.9	7.13	4.93	0.71	0.66	272.4	1.89	4.58	0.799	0.600
WebSSL-1B	100K	53.87	25.5	6.57	0.61	0.59	124.0	9.59	6.37	0.756	0.515
+iREPA	100K	77.47	16.6	6.18	0.66	0.61	177.6	5.09	6.11	0.787	0.538
WebSSL-1B	400K	116.9	9.39	5.14	0.70	0.64	250.5	2.24	4.61	0.809	0.580
+iREPA	400K	130.8	7.48	4.91	0.70	0.65	271.6	1.90	4.58	0.798	0.609
PE-Core-G	100K	42.74	32.3	6.70	0.57	0.59	97.2	14.1	6.56	0.714	0.525
+iREPA	100K	75.01	18.1	6.03	0.64	0.61	176.8	5.66	6.08	0.771	0.544
PE-Core-G	400K	109.4	10.4	5.00	0.69	0.64	238.4	2.44	4.57	0.805	0.585
+iREPA	400K	132.0	7.78	5.02	0.70	0.65	275.4	1.93	4.59	0.796	0.606

Table 4: **Contribution 3 across encoders, with and without classifier-free guidance.** iREPA improves generation quality for every encoder under both settings; SiT-XL/2 is the base model. See Appendix E for detailed results on SiT-B/2 & SiT-L/2 across all encoders.

REPRODUCIBILITY STATEMENT

We provide all implementation details and hyperparameters in Appendix G. Our code, model checkpoints, and analysis results will be released upon acceptance.

REFERENCES

- Kristian Abstreiter, Sarthak Mittal, Stefan Bauer, Bernhard Schölkopf, and Arash Mehrjou. Diffusion-based representation learning. In *ICML 2021 Workshop on Unsupervised Reinforcement Learning*, 2021.
- Stability AI. Improved autoencoders ... <https://huggingface.co/stabilityai/sd-vae-ft-mse>, n.d. Accessed: April 11, 2025.
- Anthropic. Claude code. <https://claude.ai/code>, 2025.
- Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 15619–15629, 2023.
- Fan Bao, Shen Nie, Kaiwen Xue, Chongxuan Li, Shi Pu, Yaole Wang, Gang Yue, Yue Cao, Hang Su, and Jun Zhu. All are worth words: A vit backbone for diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 22669–22679, 2023.

-
- Herbert Bay, Tinne Tuytelaars, and Luc Van Gool. Surf: Speeded up robust features. In *Computer Vision—ECCV 2006: 9th European Conference on Computer Vision, Graz, Austria, May 7–13, 2006. Proceedings, Part I 9*, pp. 404–417. Springer, 2006.
- Daniel Bolya, Po-Yao Huang, Peize Sun, Jang Hyun Cho, Andrea Madotto, Chen Wei, Tengyu Ma, Jiale Zhi, Jathushan Rajasegaran, Hanoona Rasheed, Junke Wang, Marco Monteiro, Hu Xu, Shiyu Dong, Nikhila Ravi, Daniel Li, Piotr Dollár, and Christoph Feichtenhofer. Perception encoder: The best visual embeddings are not at the output of the network. *arXiv:2504.13181*, 2025.
- Shoufa Chen, Mengmeng Xu, Jiawei Ren, Yuren Cong, Sen He, Yanping Xie, Nicu Sebe, and Wai Lam. Gentron: Diffusion transformers for image and video generation. *arXiv preprint arXiv:2312.04557*, 2023.
- Shoufa Chen, Chongjian Ge, Shilong Zhang, Peize Sun, and Ping Luo. Pixelflow: Pixel-space generative models with flow. *arXiv preprint arXiv:2504.07963*, 2025.
- Xinlei Chen, Saining Xie, and Kaiming He. An empirical study of training self-supervised vision transformers. In *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 9640–9649, 2021.
- Xinlei Chen, Zhuang Liu, Saining Xie, and Kaiming He. Deconstructing denoising diffusion models for self-supervised learning. *arXiv preprint arXiv:2401.14404*, 2024.
- Jang Hyun Cho, Andrea Madotto, Effrosyni Mavroudi, Triantafyllos Afouras, Tushar Nagarajan, Muhammad Maaz, Yale Song, Tengyu Ma, Shuming Hu, Hanoona Rasheed, Peize Sun, Po-Yao Huang, Daniel Bolya, Suyog Jain, Miguel Martin, Huiyu Wang, Nikhila Ravi, Shashank Jain, Temmy Stark, Shane Moon, Babak Damavandi, Vivian Lee, Andrew Westbury, Salman Khan, Philipp Krähenbühl, Piotr Dollár, Lorenzo Torresani, Kristen Grauman, and Christoph Feichtenhofer. Perceptionlm: Open-access data and models for detailed visual understanding. *arXiv:2504.13180*, 2025.
- N. Dalal and B. Triggs. Histograms of oriented gradients for human detection. In *2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR’05)*, volume 1, pp. 886–893 vol. 1, 2005. doi: 10.1109/CVPR.2005.177.
- Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*, pp. 248–255. Ieee, 2009.
- Prafulla Dhariwal and Alexander Nichol. Diffusion models beat gans on image synthesis. *Advances in neural information processing systems*, 34:8780–8794, 2021.
- Patrick Esser, Sumith Kulal, Andreas Blattmann, Rahim Entezari, Jonas Müller, Harry Saini, Yam Levi, Dominik Lorenz, Axel Sauer, Frederic Boesel, et al. Scaling rectified flow transformers for high-resolution image synthesis. In *Forty-first International Conference on Machine Learning*, 2024.
- David Fan, Shengbang Tong, Jiachen Zhu, Koustuv Sinha, Zhuang Liu, Xinlei Chen, Michael Rabbat, Nicolas Ballas, Yann LeCun, Amir Bar, et al. Scaling language-free visual representation learning. *arXiv preprint arXiv:2504.01017*, 2025.
- Michael Fuest, Pingchuan Ma, Ming Gui, Johannes Schusterbauer, Vincent Tao Hu, and Bjorn Ommer. Diffusion models and representation learning: A survey. *arXiv preprint arXiv:2407.00783*, 2024.
- Zhengyang Geng, Mingyang Deng, Xingjian Bai, J Zico Kolter, and Kaiming He. Mean flows for one-step generative modeling. *arXiv preprint arXiv:2505.13447*, 2025.
- Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial networks. *Communications of the ACM*, 63(11):139–144, 2020.

- Greg Heinrich, Mike Ranzinger, Hongxu, Yin, Yao Lu, Jan Kautz, Andrew Tao, Bryan Catanzaro, and Pavlo Molchanov. Radiov2.5: Improved baselines for agglomerative vision foundation models, 2024. URL <https://arxiv.org/abs/2412.07679>.
- Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *Advances in neural information processing systems*, 30, 2017.
- Jing Huang, S.R. Kumar, M. Mitra, Wei-Jing Zhu, and R. Zabih. Image indexing using color correlograms. In *Proceedings of IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, pp. 762–768, 1997. doi: 10.1109/CVPR.1997.609412.
- Drew A Hudson, Daniel Zoran, Mateusz Malinowski, Andrew K Lampinen, Andrew Jaegle, James L McClelland, Loic Matthey, Felix Hill, and Alexander Lerchner. Soda: Bottleneck diffusion models for representation learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 19461–19471, 2024.
- Sadeep Jayasumana, Srikumar Ramalingam, Andreas Veit, Daniel Glasner, Ayan Chakrabarti, and Sanjiv Kumar. Rethinking fid: Towards a better evaluation metric for image generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 9307–9315, June 2024.
- Dengyang Jiang, Mengmeng Wang, Liuzhuozheng Li, Lei Zhang, Haoyu Wang, Wei Wei, Guang Dai, Yanning Zhang, and Jingdong Wang. No other representation component is needed: Diffusion transformers can provide representation guidance by themselves. *arXiv preprint arXiv:2505.02831*, 2025.
- Jiatao Jing, Gu Wu, Yuanpeng Zhou, Yuxin Xu, Zixun Sun, Xinhao Ni, Yazhou Wang, Chunmeng Li, Hao Tang, Ruihua Chen, et al. Dart: Denoising autoregressive transformers for scalable text-to-image generation. *arXiv preprint arXiv:2410.08159*, 2024.
- Minguk Kang, Richard Zhang, Connelly Barnes Nga Tran, Saurabh Kar, and Jun-Yan Zhu. Distilling diffusion models into conditional gans. *arXiv preprint arXiv:2405.05967*, 2024.
- Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.
- Theodoros Kouzelis, Efstathios Karypidis, Ioannis Kakogeorgiou, Spyros Gidaris, and Nikos Komodakis. Boosting generative image modeling via joint image-feature synthesis. *arXiv preprint arXiv:2504.16064*, 2025.
- Tuomas Kynkäänniemi, Tero Karras, Samuli Laine, Jaakko Lehtinen, and Timo Aila. Improved precision and recall metric for assessing generative models. *Advances in neural information processing systems*, 32, 2019.
- Tuomas Kynkäänniemi, Tero Karras, Miika Aittala, Timo Aila, and Jaakko Lehtinen. The role of imagenet classes in fréchet inception distance. In *International Conference on Learning Representations*, 2023.
- Xingjian Leng, Jaskirat Singh, Yunzhong Hou, Zhenchang Xing, Saining Xie, and Liang Zheng. Repa-e: Unlocking vae for end-to-end tuning with latent diffusion transformers. *arXiv preprint arXiv:2504.10483*, 2025a.
- Xingjian Leng, Jaskirat Singh, Ryan Murdock, Ethan Smith, Rebecca Li, Yunzhong Hou, Zhenchang Xing, Saining Xie, and Liang Zheng. Family of End-to-End Tuned VAEs for Supercharging T2I Diffusion Transformers. <https://end2end-diffusion.github.io/repa-e-t2i/>, 2025b.
- Tianhong Li and Kaiming He. Back to basics: Let denoising generative models denoise. *arXiv preprint arXiv:2511.13720*, 2025a.
- Tianhong Li and Kaiming He. Back to basics: Let denoising generative models denoise. *arXiv preprint arXiv:2511.13720*, 2025b.

- I Loshchilov. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017.
- David G Lowe. Object recognition from local scale-invariant features. In *Proceedings of the seventh IEEE international conference on computer vision*, volume 2, pp. 1150–1157. Ieee, 1999.
- Nanye Ma, Mark Goldstein, Michael S Albergo, Nicholas M Boffi, Eric Vanden-Eijnden, and Saining Xie. Sit: Exploring flow and diffusion-based generative models with scalable interpolant transformers. In *European Conference on Computer Vision*, pp. 23–40. Springer, 2024.
- Yiyang Ma, Xingchao Liu, Xiaokang Chen, Wen Liu, Chengyue Wu, Zhiyu Wu, Zizheng Pan, Zhenda Xie, Haowei Zhang, Xingkai Yu, et al. Janusflow: Harmonizing autoregression and rectified flow for unified multimodal understanding and generation. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 7739–7751, 2025a.
- Zehong Ma, Longhui Wei, Shuai Wang, Shiliang Zhang, and Qi Tian. Deco: Frequency-decoupled pixel diffusion for end-to-end image generation. *arXiv preprint arXiv:2511.19365*, 2025b.
- Charlie Nash, Jacob Menick, Sander Dieleman, and Peter Battaglia. Generating images with sparse representations. In *International Conference on Machine Learning*, pp. 7958–7968. PMLR, 2021.
- OpenAI. Gpt-5. <https://openai.com>, 2025.
- Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervision. *Transactions on Machine Learning Research Journal*, pp. 1–31, 2024.
- Sayak Paul. Cmmd-pytorch: Pytorch implementation of cmmd metric. <https://github.com/sayakpaul/cmmd-pytorch>, 2024. Accessed: 2025.
- William Peebles and Saining Xie. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 4195–4205, 2023.
- Pablo Pernias, Dominic Rampas, Mats Leon Richter, Christopher Pal, and Marc Aubreville. Würstchen: An efficient architecture for large-scale text-to-image diffusion models. In *The Twelfth International Conference on Learning Representations*, 2023.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pp. 8748–8763. PMLR, 2021.
- Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, Eric Mintun, Junting Pan, Kalyan Vasudev Alwala, Nicolas Carion, Chao-Yuan Wu, Ross Girshick, Piotr Dollár, and Christoph Feichtenhofer. Sam 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024. URL <https://arxiv.org/abs/2408.00714>.
- Ethan Rublee, Vincent Rabaud, Kurt Konolige, and Gary Bradski. Orb: An efficient alternative to sift or surf. In *2011 International conference on computer vision*, pp. 2564–2571. Ieee, 2011.
- Tim Salimans, Ian Goodfellow, Wojciech Zaremba, Vicki Cheung, Alec Radford, and Xi Chen. Improved techniques for training gans. *Advances in neural information processing systems*, 29, 2016.
- Axel Sauer, Katja Schwarz, and Andreas Geiger. Projected gans converge faster. In *Advances in Neural Information Processing Systems*, volume 34, pp. 17480–17492, 2021.
- Axel Sauer, Katja Schwarz, and Andreas Geiger. Stylegan-xl: Scaling stylegan to large diverse datasets. In *ACM SIGGRAPH 2022 conference proceedings*, pp. 1–10, 2022.
- Axel Sauer, Tero Karras, Samuli Laine, Andreas Geiger, and Timo Aila. Stylegan-t: Unlocking the power of gans for fast large-scale text-to-image synthesis. *arXiv preprint arXiv:2301.09515*, 2023.

- Axel Sauer, Dominik Lorenz, Andreas Blattmann, and Robin Rombach. Adversarial diffusion distillation. In *European Conference on Computer Vision*, pp. 218–234. Springer, 2024.
- Eli Shechtman and Michal Irani. Matching local self-similarities across images and videos. In *2007 IEEE Conference on Computer Vision and Pattern Recognition*, pp. 1–8, 2007. doi: 10.1109/CVPR.2007.383198.
- Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, Francisco Massa, Daniel Haziza, Luca Wehrstedt, Jianyuan Wang, Timothée Darcet, Théo Moutakanni, Leonel Sentana, Claire Roberts, Andrea Vedaldi, Jamie Tolan, John Brandt, Camille Couprie, Julien Mairal, Hervé Jégou, Patrick Labatut, and Piotr Bojanowski. DINOv3, 2025. URL <https://arxiv.org/abs/2508.10104>.
- Karen Simonyan and Andrew Zisserman. Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556*, 2014.
- George Stein, Jesse C. Cresswell, Rasa Hosseinzadeh, Yi Sui, Brendan Leigh Ross, Valentin Villecroze, Zhaoyan Liu, Anthony L. Caterini, J. Eric T. Taylor, and Gabriel Loaiza-Ganem. Exposing flaws of generative model evaluation metrics and their unfair treatment of diffusion models. In *Advances in Neural Information Processing Systems*, 2023.
- Shitao Tang, Fuyang Chen, Aviral Joshi Bose, Kun Zhang, Shengjie Bi, Zhenyu Wang, Jiacheng Feng, Haozhe Cai, Alexander Liu, Antonio Torralba, et al. Ardit: Autoregressive diffusion transformers. *arXiv preprint arXiv:2406.00843*, 2024.
- Dmitry Ulyanov, Andrea Vedaldi, and Victor Lempitsky. Instance normalization: The missing ingredient for fast stylization. *arXiv preprint arXiv:1607.08022*, 2016.
- Runqian Wang and Kaiming He. Diffuse and disperse: Image generation with representation regularization. *arXiv preprint arXiv:2506.09027*, 2025.
- Shuai Wang, Ziteng Gao, Chenhui Zhu, Weilin Huang, and Limin Wang. Pixnerd: Pixel neural field diffusion. *arXiv preprint arXiv:2507.23268*, 2025a.
- Shuai Wang, Zhi Tian, Weilin Huang, and Limin Wang. Ddt: Decoupled diffusion transformer. *arXiv preprint arXiv:2504.05741*, 2025b.
- Ziqiao Wang, Wangbo Zhao, Yuhao Zhou, Zekai Li, Zhiyuan Liang, Mingjia Shi, Xuanlei Zhao, Pengfei Zhou, Kaipeng Zhang, Zhangyang Wang, Kai Wang, and Yang You. Repa works until it doesn’t: Early-stopped, holistic alignment supercharges diffusion training. *arXiv preprint arXiv:2505.16792*, 2025c.
- Haoyu Wu, Diankun Wu, Tianyu He, Junliang Guo, Yang Ye, Yueqi Duan, and Jiang Bian. Geometry forcing: Marrying video diffusion and 3d representation for consistent world modeling. *arXiv preprint arXiv:2507.07982*, 2025.
- Mengping Yang, Ceyuan Yang, Yichi Zhang, Qingyan Bai, Yujun Shen, and Bo Dai. Revisiting the evaluation of image synthesis with gans. In *Advances in Neural Information Processing Systems*, 2023.
- Jingfeng Yao and Xinggang Wang. Reconstruction vs. generation: Taming optimization dilemma in latent diffusion models. *arXiv preprint arXiv:2501.01423*, 2025.
- Sihyun Yu, Sangkyung Kwak, Huiwon Jang, Jongheon Jeong, Jonathan Huang, Jinwoo Shin, and Saining Xie. Representation alignment for generation: Training diffusion transformers is easier than you think. *arXiv preprint arXiv:2410.06940*, 2024.
- Xiangdong Zhang, Jiaqi Liao, Shaofeng Zhang, Fanqing Meng, Xiangpeng Wan, Junchi Yan, and Yu Cheng. Videorepa: Learning physics for video generation through relational alignment with foundation models. *arXiv preprint arXiv:2505.23656*, 2025.

A DETAILED CORRELATION ANALYSIS

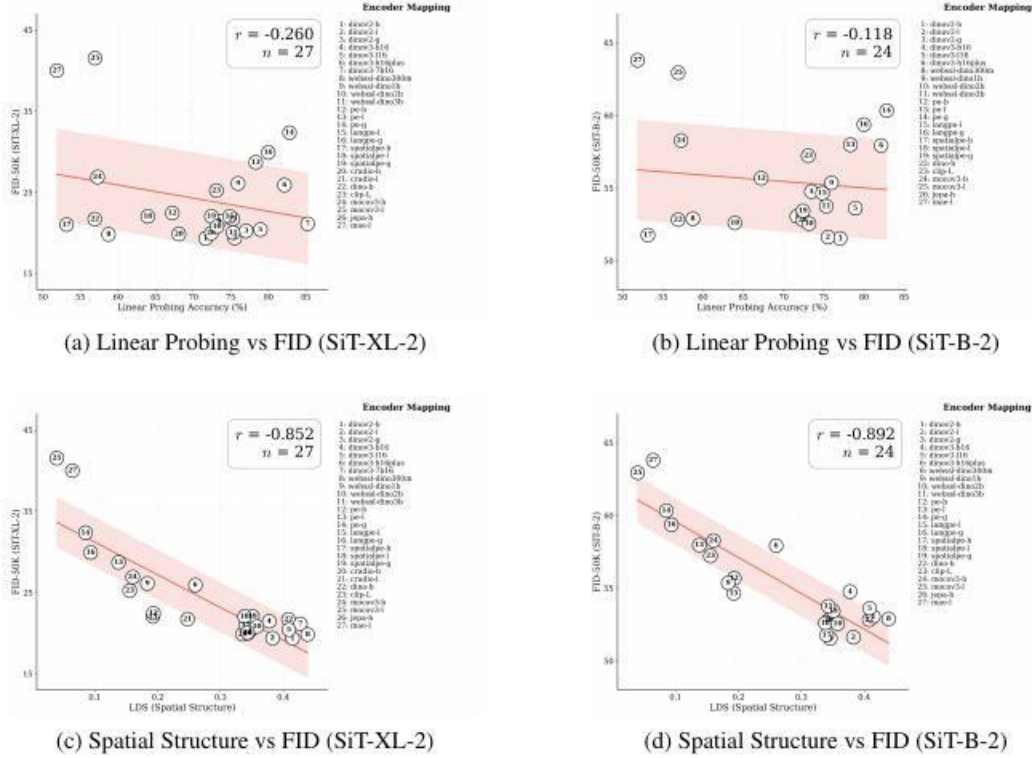


Figure 10: Detailed correlation analysis with encoder labels (Contribution 2, expanded). Top row: linear probing accuracy vs. FID for SiT-XL-2 and SiT-B-2. Bottom row: spatial structure (LDS) vs. FID. Each point is labeled with its encoder number (see legend). At both model scales, the spatial structure metric is markedly more correlated with generation quality than linear probing.

B SPATIAL SELF-SIMILARITY METRICS

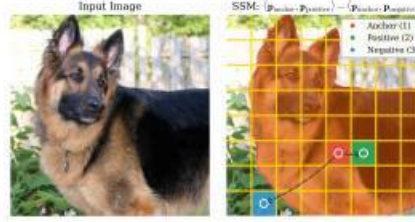


Figure 11: **Explaining semantic region self-similarity (SRSS) metric.** Visual explanation of SRSS metric: For each image, we compute a mask using SAM2 and select anchor-positive-negative triplets. The SSM measures the difference in cosine similarity between anchor-positive pairs (within mask) and anchor-negative pairs (outside mask). Intuitively, larger values indicate that patches on the same semantic region are more similar than patches on unrelated regions — indicating better spatial structure preservation.

Setup. Given an image $\mathcal{I}$ and vision encoder $\mathcal{E}$, we extract patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ with $T = H \times W$ and indices $t \in [T]$ placed on an $H \times W$ lattice. Let $d : [T] \times [T] \rightarrow \mathbb{N}$ be the Manhattan distance, and let the (cosine) self-similarity kernel be

$$K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2} \in [-1, 1],$$

a standard local self-similarity measure (Shechtman & Irani, 2007).

- *Patch token representation:* $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ where $T = H \times W$ is the spatial grid of patches (patch index set $[T]$). Also let $d : [T] \times [T] \rightarrow \mathbb{N}$ be Manhattan distance on the $H \times W$ grid.
- *Self-similarity kernel:* $K : [T] \times [T] \rightarrow \mathbb{R}$ measuring self-similarity (Shechtman & Irani, 2007) between patch tokens. We use the cosine kernel $K_{\mathcal{X}}(t, t') = \langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle / (\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2)$.
- *Spatial self-similarity metric:* a functional $m : \mathcal{K} \times \mathcal{D} \rightarrow \mathbb{R}$ measures how self-similarity K between patch tokens varies with lattice distance d . Intuitively, larger values indicate stronger spatial organization (near patches more similar to each other than far away patches).

Building on this setup, we now detail the family of straightforward metrics behind Contribution 2 for measuring the spatial self-similarity structure of a target representation. We stress that the contribution is the *signal*, not any one estimator: the four metrics below differ in form but agree in their conclusions.

Local vs. Distant Similarity (LDS). We first consider a simple correlogram contrast (local vs. distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') < r_{\text{near}}] - \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') \geq r_{\text{far}}],$$

where r_{near} and r_{far} are the hyperparameters. By default we use $r_{\text{near}} = r_{\text{far}} = H/2$. We found correlation to be robust to their exact choices. Intuitively, larger values indicate that *on average*, patches that are closer to each other are more similar than patches that are further away.

Correlation Decay Slope (CDS). Given the patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$, we compute the spatial correlogram $g_{\mathcal{X}}(\delta) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') = \delta]$ for distances $\delta \in \{1, \dots, \Delta\}$. We then fit a least-squares line $\hat{g}_{\mathcal{X}}(\delta) \approx \alpha + \beta\delta$ and define:

$$\text{CDS}(\mathcal{X}) = -\hat{\beta}$$

where $\hat{\beta}$ is the fitted slope. Larger CDS values indicate faster similarity decay with distance, suggesting stronger spatial organization.

Semantic-Region Self-Similarity (SRSS). For each image, we first sample a binary mask $\mathbf{M} \in \mathbb{R}^{H \times W}$ containing a random object using SAM2 (Ravi et al., 2024) and downsampled to $H \times W$. We then select three types of points: anchor, positive, and negative. The anchor and positive points are sampled from within the mask (should have similar semantics), while the negative point is sampled from outside the mask (less related to anchor). Conceptually, if the encoder feature representation

preserves the spatial structure, the anchor and positive points should have higher cosine similarity, while the anchor and negative points should have lower cosine similarity. Thus, we define the Spatial Structure Metric (SSM) as:

$$\text{SSM}(\mathbf{P}) = \mathbb{E}_{\text{anchor} \in \mathbf{M}} [\cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{pos}}) - \cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{neg}})]$$

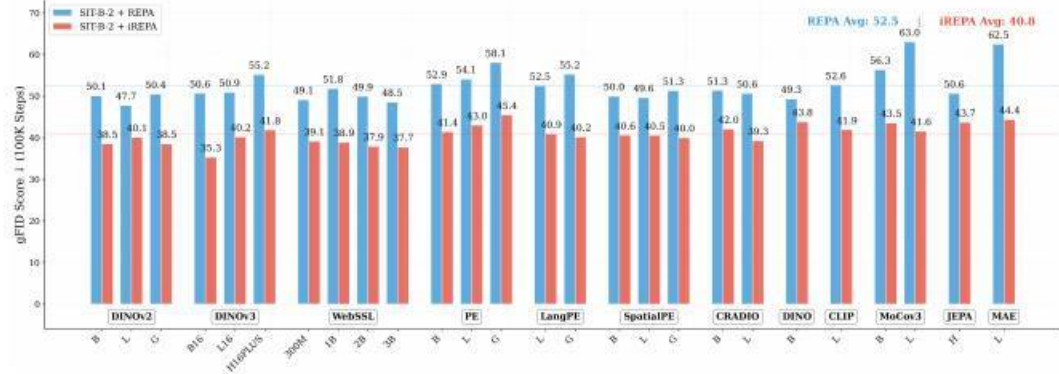
where positive points are within Manhattan distance $d \leq r_{\text{near}}$ from the anchor within the mask, and negative points are at distance $d \geq r_{\text{far}}$ outside the mask.

RMS Spatial Contrast (RMSC). Finally, we consider a simple contrast metric to capture the spatial contrast between patch token representations. Given normalized features $\hat{\mathbf{x}}_t = \mathbf{x}_t / \|\mathbf{x}_t\|_2$ for each patch t , we compute:

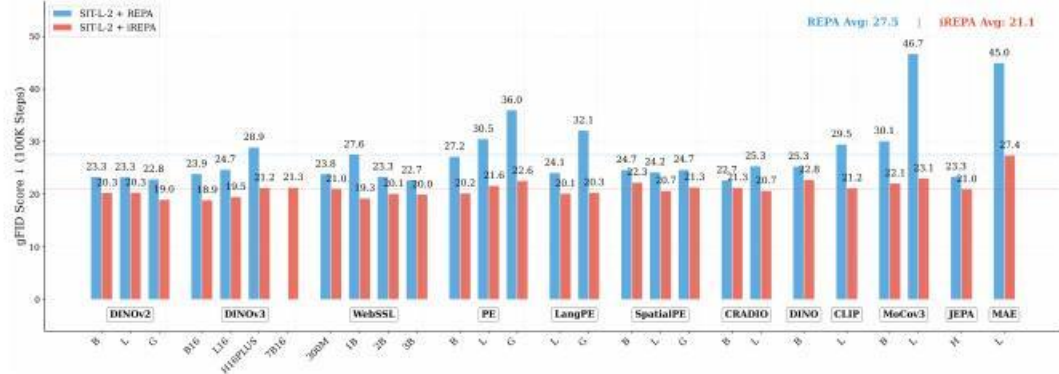
$$\text{RMSC}(\mathcal{X}) = \sqrt{\frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{x}}_t - \bar{\mathbf{x}}\|_2^2}$$

where $\bar{\mathbf{x}} = \frac{1}{T} \sum_{t=1}^T \hat{\mathbf{x}}_t$ is the mean of normalized features across all spatial locations. Higher RMSC values indicate greater spatial diversity in the feature representations, suggesting preserved spatial structure, whereas lower values indicate more uniform feature distributions indicating a loss of spatial structure (typically happens global information diminishes spatial structure (Siméoni et al., 2025)).

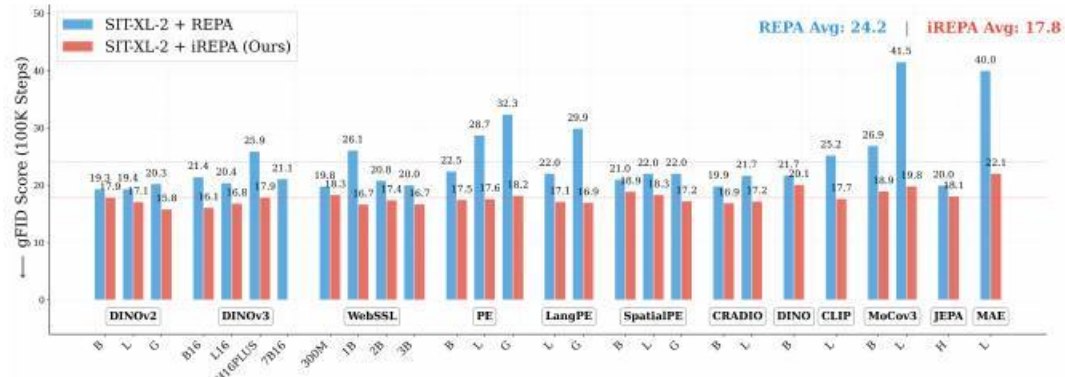
C VISION ENCODER VARIATION



(a) SiT-B-2 (130M parameters)



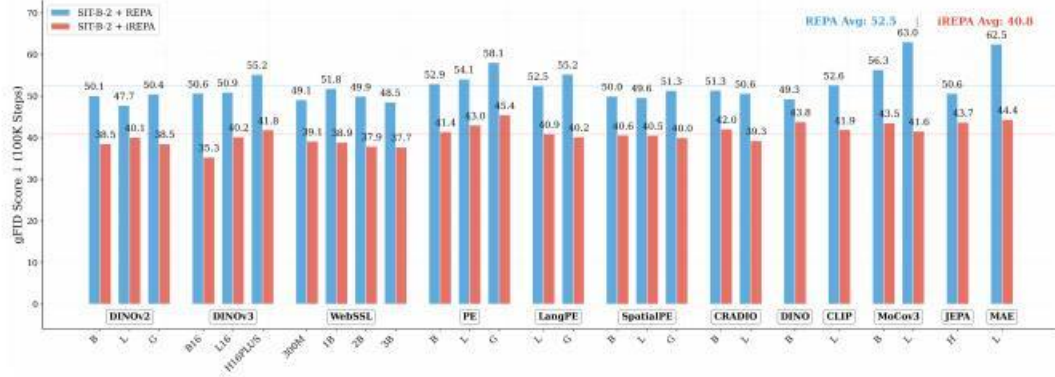
(b) SiT-L-2 (458M parameters)



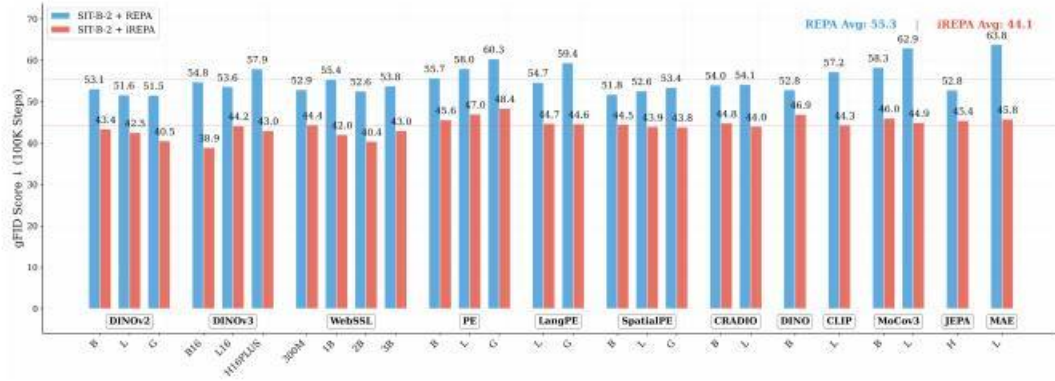
(c) SiT-XL-2 (675M parameters)

Figure 12: **Variation in vision encoders.** Ablation studies showing iREPA improvements across different vision encoders for SiT-B-2, SiT-L-2, and SiT-XL-2 models. iREPA consistently improves generation quality across all encoders and model sizes.

D ENCODER DEPTH VARIATION



(a) SiT-B-2 with 6 encoder layers



(b) SiT-B-2 with 8 encoder layers

Figure 13: **Contribution 3 is robust to encoder depth.** SiT-B-2 performance at two encoder depths (6 vs. 8 layers) across various vision encoders. iREPA improves consistently regardless of depth, with slightly stronger results at 8 layers, and both configurations show significant FID gains once spatial regularization is applied.

E COMPREHENSIVE RESULTS ACROSS VISION ENCODERS

E.1 SiT-B/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	59.44	24.84	6.46	0.584	0.645
+iREPA	69.70	21.26	6.64	0.601	0.644
DINOv2-B	64.92	22.75	6.54	0.593	0.653
+iREPA	70.88	21.40	6.77	0.597	0.653
DINOv2-L	66.65	22.46	6.53	0.594	0.650
+iREPA	72.41	20.99	6.84	0.600	0.650
DINOv3-B16	62.57	23.91	6.77	0.584	0.647
+iREPA	75.48	19.65	6.68	0.606	0.649
DINOv3-H16+	61.91	24.82	6.57	0.579	0.645
+iREPA	71.76	21.80	6.84	0.592	0.655
DINO-B	58.37	24.71	6.32	0.586	0.635
+iREPA	57.55	24.41	6.43	0.587	0.638
LangPE-G	58.42	25.37	6.32	0.583	0.641
+iREPA	66.88	21.62	6.65	0.604	0.649
MoCov3-B	57.61	25.60	6.25	0.579	0.640
+iREPA	61.41	22.80	6.46	0.596	0.639
PE-B	58.86	25.78	6.44	0.58	0.64
+iREPA	69.50	21.15	6.66	0.601	0.659
PE-G	56.09	26.84	6.45	0.568	0.641
+iREPA	70.43	21.52	6.58	0.598	0.651
SpatialPE-B	59.19	24.28	6.37	0.586	0.646
+iREPA	61.50	22.71	6.41	0.604	0.644
SpatialPE-L	62.13	23.87	6.51	0.59	0.65
+iREPA	68.83	21.07	6.59	0.607	0.651
WebSSL-1B	63.20	23.48	6.33	0.593	0.646
+iREPA	73.54	19.74	6.63	0.612	0.646
WebSSL-2B	66.56	22.27	6.69	0.596	0.654
+iREPA	72.13	20.47	6.89	0.609	0.648
WebSSL-300M	62.06	23.64	6.48	0.590	0.646
+iREPA	66.83	21.46	6.71	0.606	0.653
WebSSL-3B	65.91	22.55	6.53	0.598	0.654
+iREPA	71.98	20.64	6.71	0.607	0.650

Table 5: **Contribution 3 at SiT-B/2: comprehensive per-encoder results at 400K iterations.** iREPA improves generation quality across the full set of encoders. All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

E.2 SiT-L/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	107.7	10.6	5.20	0.682	0.643
+iREPA	115.8	9.41	5.15	0.689	0.653
DINOv2-B	113.3	9.61	5.11	0.684	0.653
+iREPA	113.9	9.36	5.11	0.690	0.655
DINOv3-B16	115.3	9.41	5.24	0.687	0.651
+iREPA	118.1	9.06	5.25	0.685	0.650
DINOv3-H16+	110.7	10.6	5.32	0.677	0.652
+iREPA	120.5	9.64	5.29	0.681	0.663
LangPE-G	102.8	11.2	5.13	0.684	0.637
+iREPA	112.7	9.37	5.02	0.690	0.653
MoCov3-B	102.1	11.0	5.14	0.685	0.635
+iREPA	104.7	10.1	5.00	0.690	0.641
PE-G	98.38	12.6	5.31	0.670	0.646
+iREPA	117.1	9.65	5.33	0.683	0.651
WebSSL-1B	107.9	10.5	5.13	0.686	0.644
+iREPA	121.5	8.62	5.01	0.696	0.653

Table 6: **Contribution 3 at SiT-L/2: comprehensive per-encoder results at 400K iterations.** iREPA improves generation quality across all encoders, with particularly strong FID and IS gains. All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

F ADDITIONAL EXPERIMENTAL RESULTS

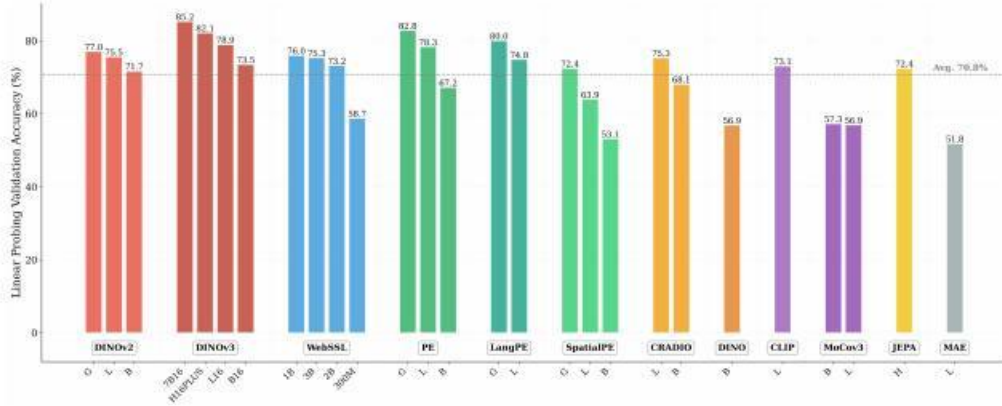


Figure 14: **Global information in mean patch tokens.** We find that in addition to [CLS] token, mean of patch tokens also contains substantial global semantic information. While this helps improve global performance, it reduces the contrast between individual patch tokens, potentially hindering spatial structure transfer. We find that we can remove some of this global information (through mean of patch tokens) to improve spatial structure transfer during representation alignment (§4).

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	262.97	1.98	4.60	0.799	0.610
+iREPA (Ours)	268.79	1.93	4.59	0.799	0.600
DINOv3-B	261.18	1.99	4.58	0.799	0.609
+iREPA (Ours)	272.41	1.89	4.58	0.799	0.600
WebSSL-1B	250.53	2.24	4.61	0.809	0.580
+iREPA (Ours)	271.59	1.90	4.58	0.798	0.609
PE-G	238.37	2.44	4.57	0.805	0.585
+iREPA (Ours)	275.36	1.93	4.59	0.796	0.606

Table 7: **Contribution 3 under classifier-free guidance.** REPA vs. iREPA with CFG (scale 2.0) across different vision encoders. All experiments use SiT-XL/2 trained for 400K iterations with 250 sampling steps. iREPA improves both IS and FID for every encoder, confirming the gains persist with guidance.

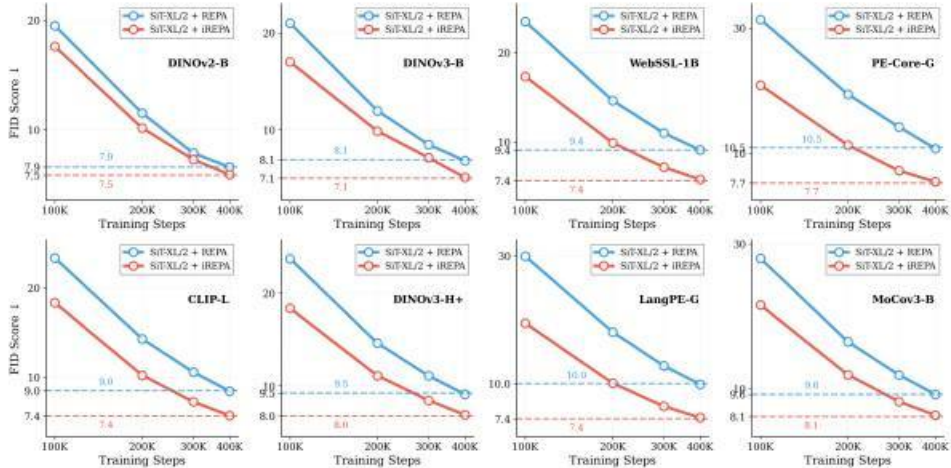


Figure 15: **Contribution 3, additional convergence curves: accentuating spatial features consistently speeds up convergence.** Results for SiT-XL/2 with REPA and iREPA.

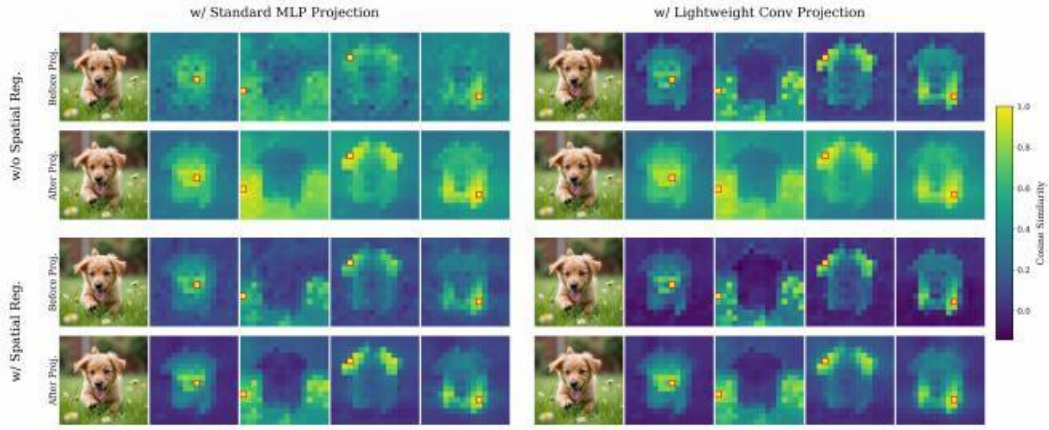


Figure 16: **Visualizing the impact of two straightforward improvements to enhance spatial feature transfer.** **First**, we find that standard MLP projection layer (top-left) losses spatial information while transferring features from the pretrained encoder (after projection) to diffusion model (before projection). Instead using a simpler convolution layer better preserves the spatial information transfer (top-right). **Second**, we observe that vision encoder features often have limited spatial contrast (§2. This causes the tokens in one semantic region (*e.g.*, dog) to show quite decent cosine similarity with unrelated tokens (*e.g.*, background). We address this with a simple spatial regularization layer, which accentuates the spatial contrast in the learned representation — leading to better generation performance. Best results are obtained while using both (refer Table 2).

G IMPLEMENTATION DETAILS

Training setup. For reproducibility of Contribution 3, we follow the same setup as REPA (Yu et al., 2024) unless otherwise specified, so that any difference is attributable to our two modifications rather than the training pipeline. All training is conducted on the ImageNet (Deng et al., 2009) training split. For preprocessing, we adopt the protocol from ADM (Dhariwal & Nichol, 2021), where each image is center-cropped and resized to 256×256 resolution. We use `stabilityai/sd-vae-ft-mse` VAE (AI, n.d.) throughout our diffusion training and inference. For spatial normalization layer we use $\gamma \in [0.6, 0.8]$ and for projection layer we use a convolutional layer with kernel size 3 and padding 1. For optimization, we use AdamW (Kingma & Ba, 2014; Loshchilov, 2017) with a constant learning rate of 1×10^{-4} , and a global batch size of 256. During training, we use `bfloat16` mixed precision and `torch.compile` to accelerate training, and gradient clipping and exponential moving average (EMA) to the generative model for stable optimization.

For REPA-E (Leng et al., 2025a) and JiT (Li & He, 2025a) experiments, we use the official open-source implementation. For MeanFlow (Geng et al., 2025) experiments, we received the implementation (including REPA) from original authors and adapt it to introduce the two straightforward changes for iREPA (§4).

Evaluation. For image generation evaluation, we strictly follow the ADM setup (Dhariwal & Nichol, 2021). We report generation quality using Fréchet inception distance (gFID) (Heusel et al., 2017), structural FID (sFID) (Nash et al., 2021), inception score (IS) (Salimans et al., 2016), precision (Prec.) and recall (Rec.) (Kynkäänniemi et al., 2019), measured on 50K generated images. For sampling, we follow the approach in REPA (Yu et al., 2024), using the SDE Euler-Maruyama sampler with 250 steps. For JiT (Li & He, 2025a), we use 50 inference steps following official implementation.

H MORE DISCUSSION ON RELATED WORK

Classical spatial features in computer vision. Spatial feature extraction has long been fundamental to computer vision. Classical methods like SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), SURF (Bay et al., 2006), and ORB (Rublee et al., 2011) provide robust local descriptors for tasks ranging from object detection to image matching. While these handcrafted features excel at capturing local spatial patterns and geometric invariances, recent generative modeling has focused on representations from modern self-supervised methods with strong global classification performance, such as DINOv2 (Oquab et al., 2024) and CLIP (Radford et al., 2021). Our findings (Contributions 1–2) invert the usual selection criterion: because spatial structure preservation is what matters for generation quality, even classical spatial features can improve diffusion training when properly aligned. This opens up the full spectrum of spatial feature extractors, from handcrafted descriptors to modern learned representations, as candidate alignment targets, provided they maintain strong spatial coherence.

Pretrained visual encoders for generative models. Pretrained visual encoders have supported generative models in several capacities: as discriminators to accelerate GAN convergence (Goodfellow et al., 2020; Sauer et al., 2021; 2022; 2023; Radford et al., 2021), as teachers in adversarial distillation for diffusion models (Sauer et al., 2024; Kang et al., 2024), and more recently as alignment targets. In GANs (Goodfellow et al., 2020), pretrained features have improved both convergence speed and scaling to large datasets, as shown by StyleGAN-XL (Sauer et al., 2022) and StyleGAN-T (Sauer et al., 2023) with CLIP (Radford et al., 2021) features. For diffusion models, adversarial distillation uses pretrained encoders to guide students toward higher-fidelity samples. In particular, REPA (Yu et al., 2024) aligns diffusion features with external encoders, showing that representation alignment improves both convergence and quality. Our contribution to this line is to identify *which* property of the encoder matters — spatial structure preservation, not discriminative capability — and to exploit it directly (Contribution 3).

Denosing transformers. Transformer architectures have become the dominant backbone for scalable generative modeling, with formulations including diffusion transformers (Peebles & Xie, 2023) and flow matching variants (Ma et al., 2024). Recent architectures like GenTron (Chen et al., 2023) scale transformers past 3B parameters for text-to-image synthesis; U-ViT (Bao et al., 2023) shows that pure transformer backbones without U-Net inductive biases can be competitive; and ARDiT (Tang et al., 2024) and DART (Jing et al., 2024) explore autoregressive formulations that trade off quality and

speed. Despite these advances, training such models from scratch remains expensive, often requiring millions of iterations. Representation alignment methods like REPA (Yu et al., 2024) dramatically accelerate convergence, but the mechanism stayed unclear — precisely the gap our first contribution fills. By showing that the key benefit comes from preserving spatial structure rather than semantic alignment, our analysis explains why some encoders accelerate training more than others and directly guides the design of the more effective alignment strategy in Contribution 3. While (Wang et al., 2025c) propose improvements to REPA training through early stopping, they hypothesize that “REPA predominantly distills global semantic information while leaving structural information untapped.” In contrast, our analysis reveals that spatial structure (not global semantic information) already plays a very significant role in the effectiveness of REPA (§2, §3). Thus, while we indeed find that spatial structure remains underexploited (§4) in REPA, we surprisingly find that majority of improvements in REPA are already coming from introducing an inductive bias for spatial structure (not global information).

Denosing as self-supervised learning task. The connection between denoising and representation learning has been explored from multiple perspectives. Early work by Abstreiter et al. (2021) extended diffusion objectives for representation learning, demonstrating that denoising naturally learns meaningful features. SODA (Hudson et al., 2024) introduces a diffusion model with an information bottleneck to learn compact representations. Chen et al. (2024) deconstruct diffusion models and find that a simple denoising autoencoder suffices for strong self-supervised performance. Wang & He (2025) introduce a dispersive loss to encourage internal representation diversity in diffusion models, improving generative performance without external encoders. Similarly, Jiang et al. (2025) propose Self-Representation Alignment (SRA) to align a diffusion transformer’s latent features across noise levels, providing self-guidance without an auxiliary model. These works establish denoising as a fundamental self-supervised task that naturally encourages robust features. Our findings complement this view and sharpen it: aligning diffusion models with strong *spatial* representations improves both generative and discriminative capabilities, identifying spatial structure preservation as the key factor in this synergy (Contributions 1–2).

Scaling self-supervised vision encoders. Recent years have seen remarkable progress in scaling self-supervised vision models to unprecedented sizes and datasets. DINOv3 (Siméoni et al., 2025) trains a 7B parameter ViT on 1.7 billion images without labels by aligning representations from different augmentations. WebSSL (Fan et al., 2025) demonstrates that visual models trained on more than 2 billion images can match language-supervised models like CLIP (Radford et al., 2021) on vision-language tasks without language supervision. C-RADIO (Heinrich et al., 2024) combines multiple teacher models through distillation to create versatile encoders that excel across diverse visual domains. I-JEPA (Assran et al., 2023) explores predictive architectures that learn by predicting masked regions in representation space, instead of reconstructing pixels directly. SAM (Ravi et al., 2024) specializes in promptable segmentation through large-scale supervised training. The Perception Encoder family (Bolya et al., 2025; Cho et al., 2025) shows that intermediate features often outperform final representations for dense prediction tasks. These models are typically evaluated on global tasks like image classification, but our first contribution reveals a crucial caveat: strong discriminative-benchmark performance does not translate into better generation quality. What benefits diffusion training most is spatial structure preservation, regardless of classification accuracy. A practical implication follows: the evaluation of self-supervised encoders for generative applications should weigh spatial coherence as heavily as semantic understanding.

I NOTE ON LLM USAGE

We use GPT-5 (OpenAI, 2025) for considering different variations of the spatial structure metrics discussed in the paper §3. Furthermore, all figures in the paper are directly generated from our experiment logs and checkpoints using Claude-Code (Anthropic, 2025). Additionally we use LLM help for searching and formulating relevant work in Appendix H. We use cursor in some parts to help with paper writing.

J ADDITIONAL DISCUSSION ON SPATIAL STRUCTURE METRICS

A further use of the metric (Contribution 2): spatial structure metrics can measure the representation gap. Yu et al. (2024) use linear probing to measure the representation gap, invoking the increase in validation accuracy to explain REPA’s effectiveness. We show that our spatial structure metric (SSM) offers a complementary lens: it too measures the representation gap and explains the improvements across models. As Figure 17 shows, representation alignment closes the gap between the SSM of a pretrained encoder such as DINOv2 and that of the diffusion features.

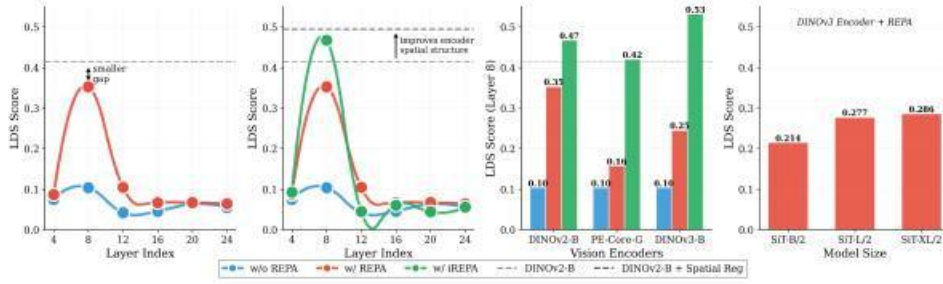


Figure 17: **Representation alignment as bridging the spatial feature gap.** a) We find that representation alignment can also be seen as bridging the spatial feature gap between diffusion and vision encoder patch features. b) iREPA (§4) accentuates the spatial features of the vision encoder (at cost of some global information) — helping achieve better generation performance. c) iREPA helps consistently improve performance across different vision encoders. d) Spatial structure improvements scale with model size. All results are reported with SiT-XL/2 at 400k iterations.

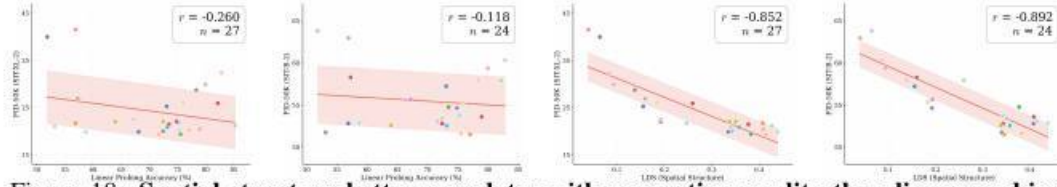


Figure 18: **Spatial structure better correlates with generation quality than linear probing.** Correlation analysis across 27 vision encoders. **Left two:** Linear probing accuracy vs FID for SiT-XL-2 (Pearson $r = -0.26$) and SiT-B-2 ($r = -0.12$) shows weak correlation. **Right two:** Spatial structure (LDS) vs FID for SiT-XL-2 ($r = -0.85$) and SiT-B-2 ($r = -0.89$) shows strong correlation. See Figure 10 in Appendix for detailed plots with encoder labels.

K ADDITIONAL ANECDOTAL COMPARISONS

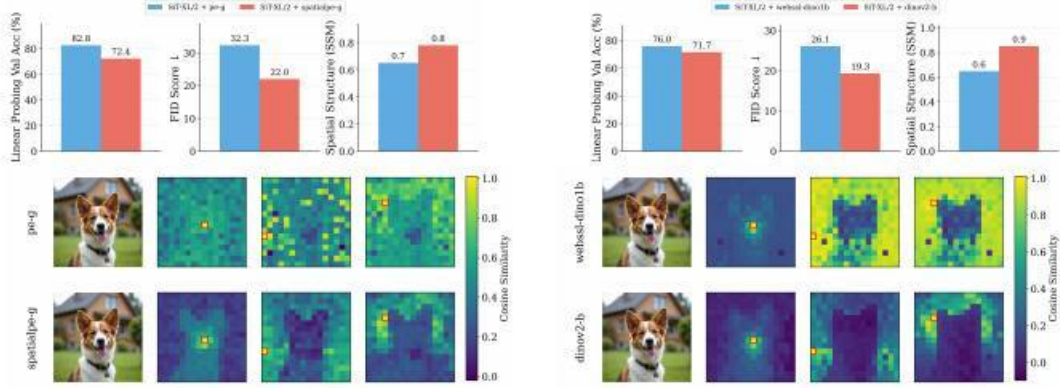


Figure 19: **Motivating anecdotes from recent SSL encoders — spatial structure matters.** **Top row:** Metrics comparison showing inverse relationship between ImageNet accuracy and generation quality. **Left:** PE-g achieves higher accuracy (82.8% vs 72.4%) but worse FID (32.3 vs 22.0) than Spatial-PE-g, with much lower spatial structure (LDS: 0.1 vs 0.4). **Right:** WebSSL-dino1b shows higher accuracy (76.0% vs 71.7%) but worse FID (26.1 vs 19.3) than DINOv2-b, with weaker spatial structure (LDS: 0.2 vs 0.4). **Bottom row:** Spatial cosine similarity visualizations confirm that encoders with better generation (Spatial-PE-g, DINOv2-b) maintain clear spatial coherence, while those optimized for classification (PE-g, WebSSL) lose spatial structure.

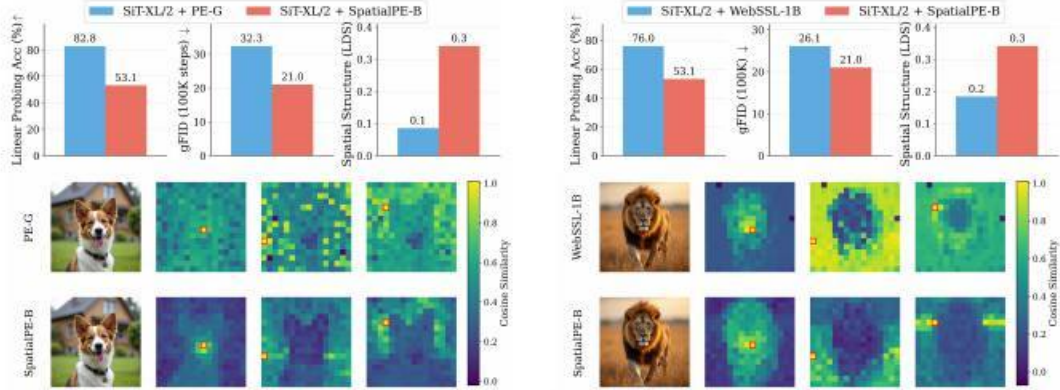


Figure 20: **Motivating examples — spatial structure matters.** **Top:** Metrics comparison showing inverse relationship between ImageNet accuracy and generation quality. **Left:** PE_{core}-G, despite having significantly higher validation accuracy (82.8% vs. 53.1%), shows worse generation quality compared to PE_{Spatial}-B (Bolya et al., 2025). **Right:** Similarly, WebSSL-1B (Fan et al., 2025) also shows much better global performance (76.0% vs. 53.1%), but worse generation. **Bottom:** We find that spatial structure instead provides a better predictor of generation quality than global performance. See §3 for spatial structure metric. All results reported at 100K using SiT-XL/2 and REPA.

L FURTHER DISCUSSION

Role of spatial normalization (why Modification 2 works). Here we visualize the mechanism behind the spatial normalization layer of Contribution 3. Vision encoders often carry a significant global component, or overlay, that limits the contrast between spatial tokens: tokens in one semantic region can be highly correlated with tokens in unrelated regions (e.g., background), blunting the spatial distinctiveness of the features. Spatial normalization removes this global overlay, sharpening the contrast between spatial tokens so the model better preserves local spatial structure while suppressing the global information that interferes with generation quality.

The following figures make this effect concrete across several examples, showing how spatial normalization reshapes the feature representations to emphasize spatial structure:

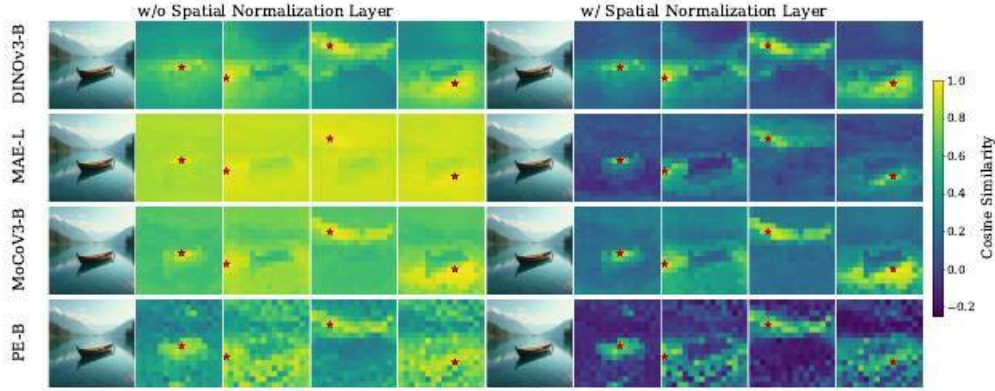


Figure 21: Visualizing impact of spatial normalization (Example 1). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while reducing global interference, resulting in more distinct semantic boundaries.

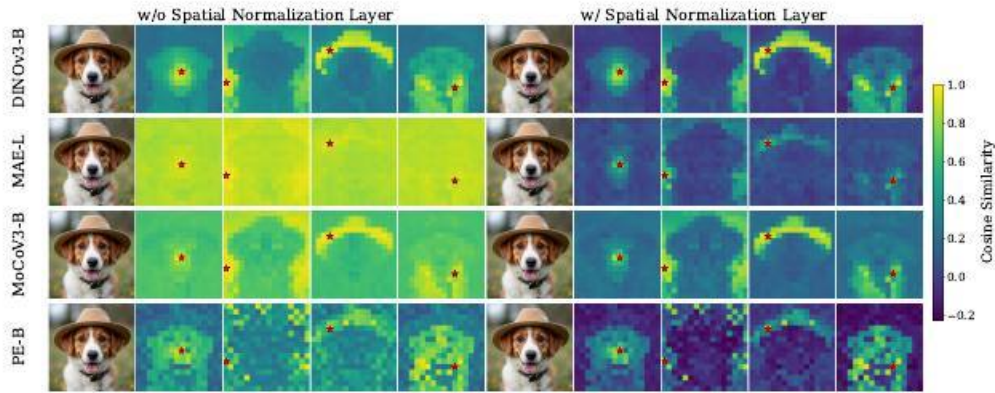


Figure 22: Visualizing impact of spatial normalization (Example 3). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while reducing global interference, resulting in more distinct semantic boundaries.

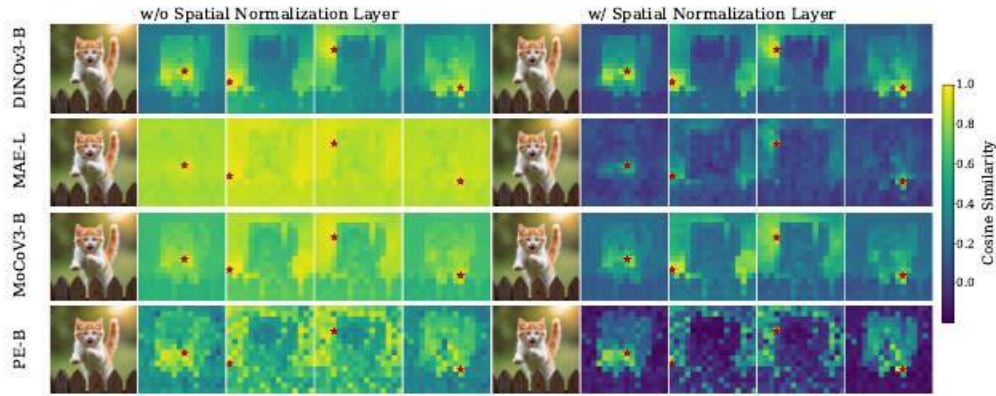


Figure 23: Visualizing impact of spatial normalization (Example 2). Feature similarity maps demonstrate how spatial normalization improves spatial contrast. The original features (left) exhibit a global overlay that reduces distinction between foreground and background regions. After normalization (right), spatial tokens become more locally coherent, with clearer separation between different semantic regions.

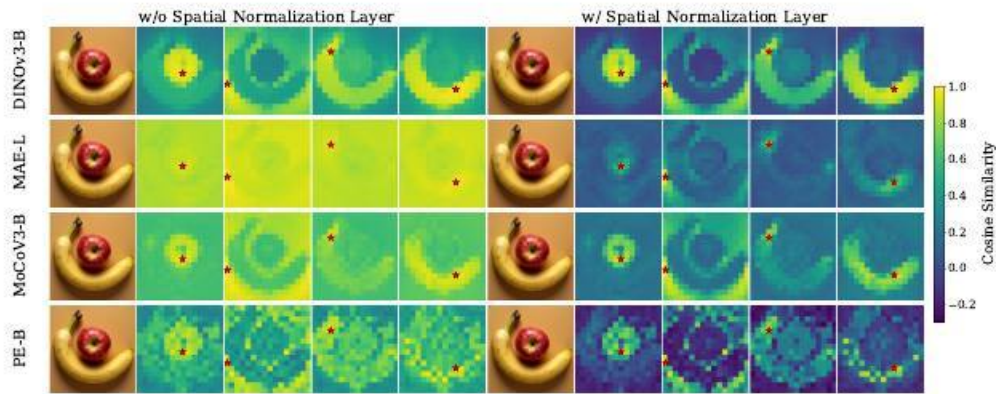


Figure 24: **Visualizing impact of spatial normalization** (Example 4). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while reducing global interference, resulting in more distinct semantic boundaries.

M ADDITIONAL RESULTS

M.1 ADDITIONAL RESULTS AT HIGHER RESOLUTIONS

Setup. To test whether Contribution 3 holds beyond ImageNet-256, we run further experiments on ImageNet-512 (Deng et al., 2009). Following the same setup as REPA (Yu et al., 2024), we report results across different pretrained encoders (DINOv2, DINOv3, WebSSL, PE, etc.), using SiT-XL and SiT-B with 50 NFE at inference and no classifier-free guidance. Fig. 25 shows REPA before and after the spatial improvements (iREPA). The gains carry over to higher resolution, and remain consistent across the choice of pretrained encoder (DINOv2, DINOv3, WebSSL, PE, etc.).

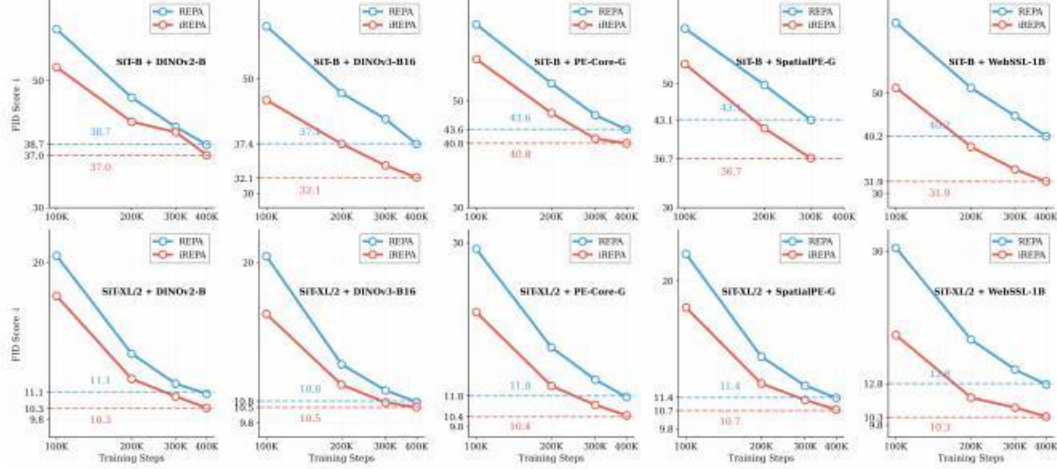


Figure 25: **Contribution 3 at 512×512 resolution.** Accentuating spatial features consistently speeds up convergence at both ImageNet 256 (Figure 7) and ImageNet 512 (above), showing the method is resolution-agnostic.

M.2 ADDITIONAL RESULTS ON MULTIMODAL T2I TASKS

Setup. To probe the generalizability of Contribution 3 beyond ImageNet class-conditional generation, we run extensive experiments on multimodal T2I tasks. Following REPA (Yu et al., 2024), we adopt MMDiT (Esser et al., 2024) as the diffusion backbone and apply REPA with various pretrained vision encoders (DINOv2, CLIP, WebSSL, PE, etc.). As in REPA (Yu et al., 2024), all models are trained for 150K steps with batch size 256 and evaluated with an ODE sampler at 50 NFE. Fig. 26 shows REPA before and after the spatial improvements (iREPA). The gains extend to the multimodal T2I setting as well, and remain consistent across the choice of pretrained encoder (DINOv2, CLIP, WebSSL, PE, etc.).

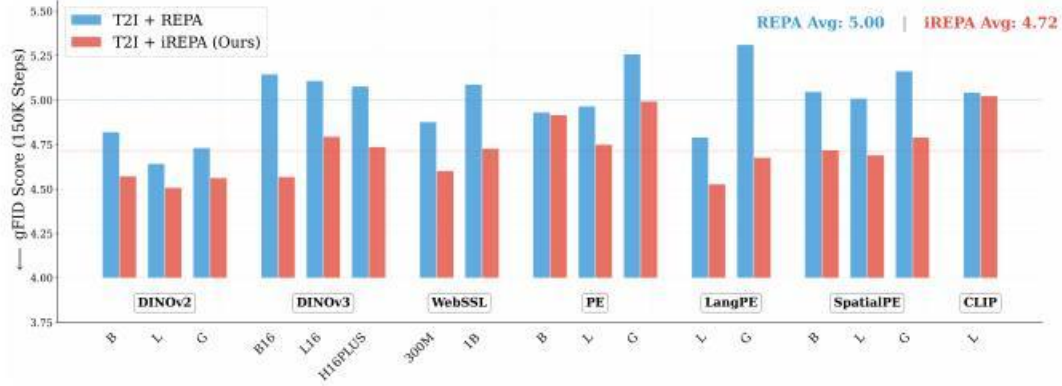


Figure 26: **Contribution 3 on text-to-image generation across encoder variants.** Accentuating the transfer of spatial features from the target representation to the diffusion features consistently speeds up convergence for both ImageNet (see §4) and multimodal T2I tasks (above), with consistent gains across pretrained encoders (DINOv2, CLIP, WebSSL, PE, etc.).

M.3 CORRELATION ANALYSIS WITHOUT OUTLIERS (MoCOv3-L AND MAE-L)

As a further robustness check on Contribution 2, we repeat the correlation analysis of Section 3 after removing the two outliers, MoCOv3-L and MAE-L. Figure 28 shows the updated correlations across model sizes. Spatial structure remains far more predictive of generation performance than linear probing accuracy. Interestingly, once the outliers are removed, linear probing even shows a small *positive* correlation with gFID (higher linear probing $\rightarrow$ worse generation). This reinforces the observations of Sec. 2 (Contribution 1): target representations with higher global semantic performance often generate similarly or worse under REPA.

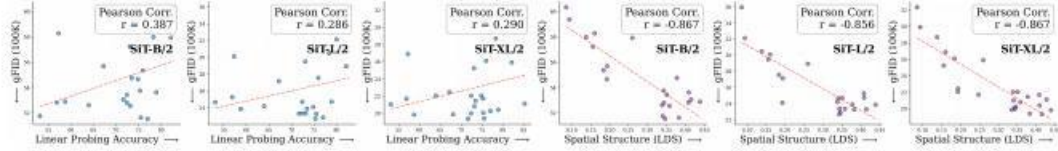


Figure 27: **Contribution 2 without outliers.** After removing the outliers (MoCOv3-L and MAE-L), spatial structure remains far more correlated (Pearson $|r| > 0.85$) with generation performance than linear probing accuracy (Pearson $|r| < 0.38$) at every model size (B, L, XL). Interestingly, once the outliers are removed, linear probing even shows a small positive correlation with gFID (higher linear probing $\rightarrow$ worse generation), consistent with the observations of Sec. 2 (Contribution 1) that higher global semantic performance often yields similar or worse generation under REPA.

M.4 ADDITIONAL QUALITATIVE RESULTS



Figure 28: **Contribution 3, qualitative view:** generation outputs before and after the spatial improvements with REPA. All results use PE-G (Bolya et al., 2025) as the pretrained vision encoder, 400K steps (80 epochs), and classifier-free guidance scale 4.0. Mirroring the quantitative gains (§4), the spatial improvements (iREPA) also visibly improve the visual quality and coherence of the generated outputs.

M.5 ADDITIONAL RESULTS WITH ALTERNATIVE EVALUATION METRICS

A natural concern is whether Contributions 2 and 3 are artifacts of the FID metric. To rule this out, we re-verify both with an alternative measure of generation quality (Stein et al., 2023; Yang et al., 2023; Kynkäänniemi et al., 2023; Jayasumana et al., 2024), the CMMD metric (Jayasumana et al., 2024), using the PyTorch implementation from (Paul, 2024) and the standard reference set from the OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021). We now check the key findings of both §3 (Contribution 2) and §4 (Contribution 3) under CMMD.

Contribution 2 under CMMD: spatial structure still correlates far more strongly than linear probing. Repeating the large-scale correlation analysis of §3 with the CMMD metric (Jayasumana et al., 2024) (Figure 29; SiT-B/2, 100K steps, REPA), every spatial metric correlates strongly with generation quality (Pearson $|r| > 0.88$) while linear probing does not (Pearson $|r| = 0.074$). The core finding of §3 (Contribution 2) thus does not depend on whether generation is measured by FID or CMMD.

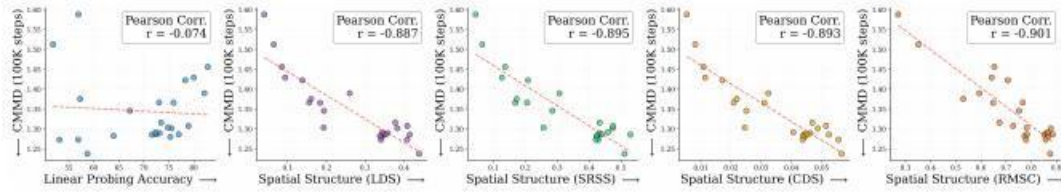


Figure 29: **Contribution 2 under the CMMD metric.** We repeat the correlation analysis of §3 with CMMD (Jayasumana et al., 2024) in place of gFID (SiT-B/2, 100K steps, REPA). Every spatial metric correlates strongly (Pearson $|r| > 0.88$) with generation performance (CMMD), while linear probing does not (Pearson $|r| = 0.074$), showing the key finding of §3 is robust to the choice of evaluation metric.

Contribution 3 under CMMD: accentuating spatial transfer still improves generation. We likewise re-check the analysis of §4 with CMMD (Jayasumana et al., 2024) alongside the traditional metrics (gFID, sFID, IS, etc.). As Table 8 shows, across vision encoders iREPA improves convergence speed under both CMMD and the traditional metrics (IS, FID, sFID, Prec., Recall), and the improvement holds both with and without classifier-free guidance (CFG).

Vision Encoder	Steps	w/o CFG						w/ CFG					
		CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑	CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑
DINOv2-B	100K	0.702	69.20	19.3	5.89	0.64	0.61	0.529	157.2	6.35	5.91	0.77	0.54
+iREPA	100K	0.652	77.92	16.9	6.26	0.66	0.61	0.484	179.3	5.15	6.23	0.78	0.54
DINOv2-B	400K	0.455	127.4	7.76	5.06	0.70	0.66	0.320	263.0	1.98	4.60	0.80	0.61
+iREPA	400K	0.438	128.6	7.52	4.89	0.71	0.65	0.310	268.8	1.93	4.59	0.80	0.60
DINOv3-B	100K	0.749	63.64	21.4	6.14	0.63	0.60	0.571	144.0	7.57	6.09	0.76	0.53
+iREPA	100K	0.651	78.79	16.2	6.14	0.66	0.61	0.481	181.9	4.87	6.10	0.78	0.55
DINOv3-B	400K	0.474	126.7	8.10	5.06	0.70	0.66	0.336	261.2	1.99	4.58	0.80	0.61
+iREPA	400K	0.441	132.9	7.13	4.93	0.71	0.66	0.314	272.4	1.89	4.58	0.80	0.60
WebSSL-1B	100K	0.825	53.87	25.5	6.57	0.61	0.59	0.622	124.0	9.59	6.37	0.76	0.52
+iREPA	100K	0.653	77.47	16.6	6.18	0.66	0.61	0.484	177.6	5.09	6.11	0.79	0.54
WebSSL-1B	400K	0.512	116.9	9.39	5.14	0.70	0.64	0.354	250.5	2.24	4.61	0.81	0.58
+iREPA	400K	0.445	130.8	7.48	4.91	0.70	0.65	0.311	271.6	1.90	4.58	0.80	0.61
PE-Core-G	100K	0.922	42.74	32.3	6.70	0.57	0.59	0.714	97.2	14.1	6.56	0.71	0.53
+iREPA	100K	0.697	75.01	18.1	6.03	0.64	0.61	0.525	176.8	5.66	6.08	0.77	0.54
PE-Core-G	400K	0.525	109.4	10.4	5.00	0.69	0.64	0.366	238.4	2.44	4.57	0.81	0.59
+iREPA	400K	0.458	132.0	7.78	5.02	0.70	0.65	0.322	275.4	1.93	4.59	0.80	0.61
CLIP-L	100K	0.790	54.07	25.2	6.65	0.61	0.60	0.605	124.5	9.77	6.59	0.75	0.52
+iREPA	100K	0.657	74.46	17.8	6.33	0.65	0.61	0.489	172.0	5.67	6.33	0.78	0.54
CLIP-L	400K	0.487	117.8	8.97	4.98	0.70	0.65	0.339	258.7	2.15	4.64	0.81	0.59
+iREPA	400K	0.442	130.1	7.42	5.03	0.71	0.65	0.307	271.8	1.96	4.64	0.80	0.61

Table 8: **Contribution 3 is robust to the evaluation metric.** We repeat the comparison with the CMMD metric (Jayasumana et al., 2024). All results use SiT-XL/2, batch size 256, and traditional REPA as baseline. We adopt the PyTorch implementation from (Paul, 2024) for CMMD, with the standard reference set from the OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021) as reference images. Across vision encoders, accentuating the transfer of spatial features (iREPA) improves convergence speed under both CMMD and the traditional metrics (IS, FID, etc.), confirming that the key findings of §4 do not depend on the choice of evaluation metric.

M.6 SPATIAL IMPROVEMENTS WITH SAM2

Table 9 and Figure 30 examine the spatial normalization component of Contribution 3 (§4) on SAM2-S (46M) encoder features². This case is a useful boundary test of the mechanism: spatial normalization works by removing the global component (mean of patch tokens) to sharpen spatial contrast, so its benefit should shrink when there is little global component to remove. That is exactly what we see — for SAM2, which already carries almost no global information (validation accuracy < 24%), spatial normalization helps only marginally (Table 9, Figure 30), consistent with the mechanism underlying Contribution 3 rather than an exception to it.

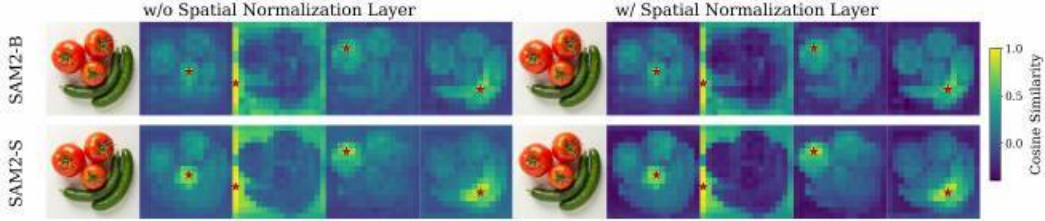


Figure 30: **Impact of spatial normalization on SAM2.** We observe that for SAM2, while use of spatial normalization layer does help enhance the spatial contrast, the improvements can be marginal. This is because spatial normalization (§4) relies on removing the global component (mean of patch tokens) to enhance spatial contrast. Since SAM2 already has little to no global information (validation accuracy < 24%), spatial normalization only slightly improves the spatial contrast.

Vision Encoder	Steps	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
SAM2-S	100K	50.69	25.32	6.52	0.631	0.588
+iREPA	100K	53.14	24.52	6.28	0.629	0.591
SAM2-S	400K	110.77	9.54	4.93	0.699	0.638
+iREPA	400K	114.62	9.10	4.89	0.704	0.640

Table 9: **Contribution 3 applied to SAM2.** Impact of the spatial improvements on the SAM2 encoder. All results are reported with SiT-XL/2, without classifier-free guidance, SAM2-S (46M) as vision encoder, and traditional REPA as the baseline.

²Note that, similar to (Bolya et al., 2025), we use the intermediate output of the vision encoder for SAM2 features rather than the mask logits. As shown in (Bolya et al., 2025), while mask logits give sharper spatial maps, they are not themselves suitable as a target representation.

M.7 ADDITIONAL RESULTS WITH FULL-FINETUNING ACCURACY INSTEAD OF LINEAR PROBING ACCURACY

Contribution 2 does not hinge on the choice of global-performance proxy. One might worry that our conclusions depend on measuring global performance via linear probing. We therefore repeat the correlation analysis of §3 using full-finetuning accuracy³ in place of linear probing (Fig. 31; SiT-XL/2, 100K steps, REPA). For full-finetuning accuracy we finetune the vision encoder end-to-end with learning rate $5e-5$, warmup ratio 0.1, for 30 epochs. The conclusion is unchanged: every spatial metric correlates far more strongly with generation performance than full-finetuning accuracy. Indeed, gFID shows a weak *positive* correlation with full-finetuning accuracy (Pearson $r = 0.317$) — higher accuracy, worse generation — once again matching the observations of §2 (Contribution 1) that stronger global semantic performance often means similar or worse generation under REPA.

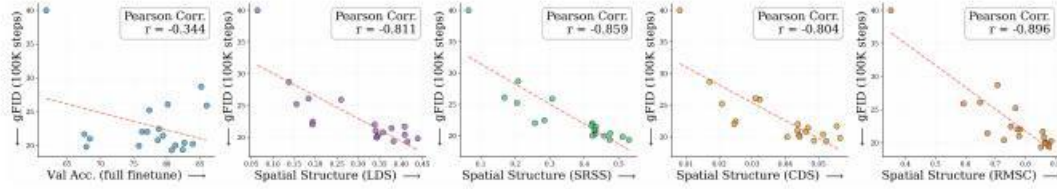


Figure 31: Contribution 2 with full-finetuning accuracy in place of linear probing. We repeat the correlation analysis of §3 using validation accuracy after full-finetuning rather than linear probing (SiT-XL/2, 100K steps, REPA). Every spatial metric remains far more correlated with generation performance than full-finetuning accuracy. Indeed, gFID shows a weak positive correlation with full-finetuning accuracy (Pearson $r = 0.317$) — higher accuracy, worse generation — consistent with §2 (Contribution 1) that higher global semantic performance often yields similar or worse generation under REPA.

³Note that, while the findings remain similar, in the context of REPA linear probing is arguably the more faithful estimate of global information in a “pretrained” encoder, since REPA regularizes against the *pretrained* features themselves; finetuning the encoder can alter how much global information it carries. As in REPA (Yu et al., 2024), we therefore keep linear probing as the default.

M.8 ADDITIONAL RESULTS ON PIXEL-SPACE DIFFUSION MODELS

Method	#Params	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
PixelFlow (Chen et al., 2025)	459M	24.67	54.33	9.71	-	0.58
PixDDT (Wang et al., 2025b)	434M	36.24	46.37	17.14	-	0.63
PixNerd (Wang et al., 2025a)	458M	43.01	37.49	10.65	-	0.62
DeCo w/o $\mathcal{L}_{\text{FreqFM}}$	426M	46.44	34.12	10.41	-	0.64
DeCo + REPA (Ma et al., 2025b)	426M	48.35	31.35	9.34	-	0.65
JiT (Li & He, 2025a)	459M	29.37	49.06	11.21	0.40	0.62
JiT+iREPA (Ours)	459M	50.72	29.19	9.42	0.51	0.65

Table 10: **Contribution 3 in pixel space: iREPA is competitive with state-of-the-art pixel-space diffusion.** Despite its simplicity, iREPA combined with JiT (Li & He, 2025a) outperforms recent pixel-space diffusion methods such as DeCo (Ma et al., 2025b). All results are reported with 200K training iterations, batch size 256, and evaluated using 50-step Euler sampling without classifier-free guidance. Results for DeCo are taken directly from (Ma et al., 2025b).